ChatGPT

Research Blueprint: Topology-Preserving 3T-to-7T MRI Enhancement for Early Alzheimer's Detection

1. Problem Re-Formulation (First Principles)

Restating the Challenge: We aim to synthesize ultra-high-field 7 Tesla (7T) brain MRI from standard 3T MRI in a way that preserves true anatomical topology and clinical biomarkers. In practical terms, this means enhancing 3T scans to **"7T-quality"** so that fine structural details (e.g. cortical layers, hippocampal subfields) become visible without introducing artifacts. The motivation is that 7T MRI offers superior resolution, signal-to-noise ratio (SNR), and gray/white matter contrast, enabling detection of subtle neurodegenerative changes (like early Alzheimer's disease pathology) that 3T often misses [1] [2]. However, 7T scanners are extremely scarce (only ~100 worldwide vs ~20,000 3T scanners [3] [4]) due to high cost and technical complexity [5] [6]. Bridging this gap via *3T-to-7T image synthesis* would democratize ultra-high-field imaging benefits for research and clinical care.

Relevance in 2024–2025: This problem remains **highly relevant and timely**. Active research in the past few years shows continued interest in 3T-to-7T MRI synthesis [7] [8]. Recent works (2020–2025) propose various deep learning approaches – from convolutional regression [9] and conditional GANs [10] to wavelet-domain CNNs [11] [12] and diffusion models [13] – indicating the field is *not saturated*. On the contrary, each new approach addresses shortcomings of previous ones (e.g. GAN artifacts, loss of fine details, limited data issues), suggesting room for genuine innovation. Furthermore, ultra-high-field MRI is gaining traction in studying diseases like Alzheimer's: 7T MRI reveals structural changes (hippocampal atrophy, microinfarcts, cortical layer thinning) that 3T cannot [1] [2]. Enabling such insights from 3T data is an unmet need in neuroimaging.

Potential Need for Reframing: We must ensure our formulation targets a *publishable novelty*. Simply doing "yet another 3T-to-7T GAN" is **incremental** now. The problem can be reframed to emphasize **anatomical fidelity and clinical validity** over purely visual super-resolution. In other words, instead of just generating sharp-looking 7T-like images, we ensure the synthetic images preserve brain topology (no spurious holes or disconnected structures) and maintain biomarker integrity (volumes, cortical thickness, hippocampal shape, etc.). This reframing addresses what reviewers and clinicians care about: *Does the synthetic 7T image lead to better neurobiological analysis?* We can also position the work as enhancing **early Alzheimer's detection** – by demonstrating that certain AD-related features (e.g. subtle hippocampal CA1 shrinkage [14] or microbleeds) become discernible on synthetic 7T images. This integration of a downstream diagnostic goal elevates the problem beyond generic image enhancement.

Why It's Worth a Q1 Paper: A successful solution would be the **first to combine ultra-high-field MRI synthesis with topology preservation and clinical validation**. It would allow widespread exploitation of 7T's benefits using existing 3T data – effectively *"software upgrading"* installed MRI scanners. This has broad impact in neuroimaging and could improve studies in Alzheimer's and other diseases by providing richer MRI data without new hardware. Moreover, our approach will contribute methodological innovations (e.g. a novel topology-aware loss or multi-task training with segmentation) that generalize to other image enhancement problems. The significance, methodological rigor, and demonstrated clinical relevance are exactly what high-tier journals seek.

Problem Scope Decisions: We will focus on **structural T1-weighted MRI** as the primary modality for synthesis, because T1w images are the clinical standard for morphology (atrophy measurement, cortical thickness) in Alzheimer's disease [1]. T2-weighted images are also available in our dataset, but will be used as needed for complementary information rather than as the main target. Specifically, our pipeline will synthesize 7T **T1w** images from 3T T1w inputs, ensuring that critical anatomical features (e.g. hippocampal subfields, cortical microstructure) are faithfully enhanced. We choose T1w as the focus over T2w or other MRI types because T1w at 7T provides the most pronounced improvements in gray–white matter contrast and small structure visibility for AD research [15] [1]. In summary, the refined research question is:

"How can we generate anatomically and clinically faithful 7T-like T1-weighted brain MR images from 3T scans, in a way that preserves topology and enhances early Alzheimer's biomarkers?"

By grounding the problem in **first principles** (MRI physics and neurobiology) and aligning it with pressing needs (access to 7T-quality data, early AD detection), we ensure it remains relevant and compelling for a Q1 audience.

2. Research Landscape & Positioning (2021–2025)

Before designing our solution, we must map the recent landscape to identify prevailing approaches and open gaps:

What Everyone Is Doing: Early works (pre-2020) on MRI up-scaling or cross-field synthesis used straightforward architectures – e.g. linear or random forest mappings [16], or basic CNN regressors minimizing MSE [17] [18]. From ~2018 onward, deep learning became dominant, with **convolutional neural networks (CNNs)** learning the non-linear 3T→7T mapping. A common baseline is a **3D U-Net** trained with voxel-wise losses (L1/L2). For example, Bahrami *et al.* 2016 applied a plain CNN with MSE loss for 3T-to-7T synthesis [19]. Soon after, generative models emerged: **GAN-based approaches** (e.g. Nie *et al.*, 2018) introduced adversarial training to better hallucinate high-frequency details [10]. GAN variants (Pix2Pix, CycleGAN if unpaired) were widely tried, as they can produce sharper, more realistic textures by learning the target distribution.

Recent Trend – Multi-Scale and Domain Fusion: Researchers realized that 3T→7T involves changes in both *resolution* and *contrast*. Recent methods incorporate multi-scale information or frequency-domain priors to handle this complexity. For instance, Qu *et al.* (2020) proposed a wavelet-based CNN (WATNet) that explicitly models low-frequency tissue contrast and high-frequency details via a wavelet transform [11] [20]. By modulating CNN feature maps with wavelet coefficients, they preserved fine anatomical details better than vanilla CNNs [21] [22]. Similarly, Zhang *et al.* (2019) introduced a dual-domain network working in image and k-space domains to enforce structural consistency [23]. These methods show a trend toward **incorporating prior knowledge (multi-resolution, physics, etc.)** into deep models to overcome the limitations of purely data-driven approaches.

Transformer and Diffusion Rise: In the 2021–2025 window, two major paradigm shifts occurred in the broader deep learning field – **Vision Transformers (ViTs)** and **Diffusion Models** – and they have started to influence medical imaging. Vision Transformers, with their attention mechanism, can capture long-range context better than CNNs, which could be beneficial for 3D MRI where global anatomical context matters. Diffusion probabilistic models (score-based generative models) have shown state-of-the-art image quality in other domains and address some GAN issues (mode collapse, training instability). Indeed, by 2024, we see the first diffusion-based 3T→7T synthesis: Li *et al.* (NeuroImage 2025) used a conditional denoising diffusion model to synthesize 7T SWI (susceptibility-weighted images) from 3T [13]. They reported that diffusion yielded **more reliable microvascular detail** than GANs, avoiding the notorious GAN training pitfalls [13]. This reflects a **rising direction**: leveraging diffusion models for MRI enhancement to improve fidelity of fine structures that GANs might distort.

Reviewer Fatigue – What's Overdone: It's clear that a generic GAN or CNN approach without a strong new twist is now *old news*. Reviewers are increasingly tired of papers that simply apply an off-the-shelf GAN (e.g. CycleGAN, Pix2Pix) to a new dataset without novel insight. By 2025, one can assume every variant of basic GAN for image translation has been seen. Especially in top venues, a method that is "just a U-Net + GAN loss + maybe a known perceptual loss" with marginal gains will be labeled **incremental**. Also, purely 2D slice-by-slice approaches are now considered suboptimal – they ignore 3D context and often result in discontinuities. Using 2D methods when 3D data is available will attract criticism for not fully utilizing spatial information [24]. Simpler CNNs that lack any integration of domain knowledge or novel learning technique might be deemed too weak. In summary, **outdated ideas** include: naive upsampling + CNN, vanilla GANs without accounting for anatomy, and any approach that focuses only on visual metrics while neglecting downstream utility.

Acceptable but Weak Directions: Some approaches are not outright wrong but may not wow reviewers. For example, a **pure supervised learning** approach on this small dataset (10 subjects) without addressing data scarcity could be seen as lacking robustness – it might work, but reviewers know it could overfit or fail on broader data. A transformer-based model could be acceptable (transformers are "modern"), but if one naively replaces a CNN with a ViT and gets similar results, reviewers might question the benefit given the complexity. Similarly, a diffusion model that *only* improves perceptual quality but doesn't tackle the core issue of anatomical accuracy might be viewed as a fancy tool applied without solving the key problems. **Weakly innovative ideas** might include using a known architecture (say, Swin-Unet or ViTGAN) for 3T-to-7T with minimal novelty beyond "we used it on a new dataset".

Genuinely Modern & Defensible Directions: To position our work strongly, we should embrace **approaches that align with current cutting-edge trends and unresolved challenges**. These include: - **Diffusion or Score-Based Generative Models:** These are hot in 2024+ and have a sound theoretical basis. Using diffusion with a clever conditioning mechanism for MRI (and possibly improving its efficiency for 3D) would be seen as novel. The recent success in SWI synthesis [8] indicates reviewers are aware of GAN limits and will appreciate a diffusion approach that demonstrably preserves more detail (e.g. veins or small hippocampal features) [25] [13]. - **Transformer-augmented Models:** Combining convolutional backbones with transformers or attention modules to capture global context is defensible. For example, a UNet where the latent features go through a self-attention block to ensure consistent global anatomy could address issues like mismatched structure shapes. However, pure ViTs can be data-hungry; a hybrid (CNN + attention) might be more suitable. If we include transformers, we need to show it helps preserve long-range structural dependencies (like symmetric structures in two hemispheres, or consistent topology). - **Multi-Modal and Multi-Task Learning:** Incorporating the **T2-weighted images** (available in our dataset) as additional inputs or outputs can set our work apart. A truly modern approach might synthesize *both* T1w and T2w at 7T from 3T inputs, leveraging the fact that multi-contrast MRI provides complementary information [26]. Even if we focus on T1w as output, using T2w 3T as an input channel in addition to T1w 3T can enhance fidelity (since T2 may help disambiguate certain tissues). Similarly, multi-task learning – for example, training the model to not only produce an image but also predict something like a segmentation or a risk score – is seen as a cutting-edge way to enforce meaningful features. Reviewers find multi-task setups appealing when they clearly improve the main task. - **Anatomy-Aware Constraints:** Introducing losses or constraints based on anatomical knowledge (topology, segmentation, surfaces) is a novel angle that directly addresses the **clinical fidelity** issue. Topology-preserving losses using concepts from persistent homology or morphological metrics are emerging in segmentation tasks and could be **first applied in image synthesis** by us [27]. For instance, TopoGAN (2020) showed that adding a topological loss (MMD between persistence diagrams of real vs fake images) improved structural faithfulness [27]. Bringing such ideas to 3D MRI synthesis would be cutting-edge. Likewise, using actual neuroanatomical segmentations (from FreeSurfer, etc.) to guide the training – e.g. ensure the synthetic image yields the same segmentation as the real 7T – is a very **2025-relevant direction** emphasizing *outcome-driven generation* [28]. - **Physics-Informed or Data-Harmonization Approaches:** An alternative modern angle is incorporating MRI physics knowledge. For example, one could enforce that the synthetic 7T image, when downsampled or filtered, should reproduce the 3T image (cycle-consistency in the sense of forward physics). Some recent works on reconstruction and enhancement have taken **physics-informed GAN** approaches that embed the MR signal model or known constraints [29] [30]. If we can encode known 7T vs 3T intensity relationships (like bias field patterns or T1 relaxation differences) into the model or training, that would be innovative. However, we must justify complexity vs payoff – any physics-based addition should clearly target a known issue (e.g. 7T images often have bias inhomogeneity; maybe our model can learn a bias field map explicitly as part of the pipeline).

Our Positioning: Considering all this, we will position our project at the intersection of **these rising directions**. We reject the tired route of a plain cGAN, and instead embrace a **diffusion-based framework with anatomy-aware enhancements** as our core novelty. We will also evaluate with **segmentation and clinical metrics**, aligning with the latest expectations that synthetic images be judged by their usefulness, not just pixel-wise similarity [31] [32]. By doing so, we ensure our approach is not only modern in method but also tuned to what reviewers are actively looking for: works that push beyond superficial image quality into maintaining *scientific utility* of the data.

In summary, our literature survey tells us to **avoid** one-dimensional GAN papers and to **adopt** a strategy that merges state-of-the-art generative modeling (diffusion, transformers) with domain-specific knowledge (MRI physics, neuroanatomy constraints). This combination – if executed well – will differentiate us in the current landscape and address the exact weaknesses that have prevented earlier 3T-to-7T works from being truly impactful (e.g. failing to preserve the hippocampus, or ignoring clinical validation) [28] [33].

3. Architecture Direction (Before Any Coding)

We now narrow down to 1–2 candidate architecture paradigms that can fulfill our goals, and we'll justify each against alternatives (GANs, naive approaches, etc.). Ultimately, we will converge on a single recommended architecture.

Candidate A: Anatomy-Guided Conditional Diffusion Model

- **Core Idea:** Use a *Conditional Denoising Diffusion Probabilistic Model (DDPM)* to generate 7T images from 3T input, augmented with anatomy guidance. In practice, this means we have a diffusion model where the **conditioning** is the 3T image (and possibly additional inputs like a tissue segmentation or second modality), and the model learns to iteratively refine pure noise into a realistic 7T image. The network architecture for the diffusion model's denoiser could be a 3D UNet, enhanced with attention mechanisms to capture long-range dependencies (e.g. a UNet with self-attention blocks or a Transformer UNet). We will incorporate a **topology/anatomy loss** during training: for example, we can feed the output (or intermediate) into a pre-trained segmentation network and penalize differences from the ground-truth segmentation, ensuring the model learns to respect organ boundaries and connectivity. Thus, the architecture is essentially *diffusion model + auxiliary anatomy constraint*. No explicit GAN discriminator is used; instead, realism comes from the diffusion probabilistic training.

- **Why It Beats GAN-Based Approaches:** Diffusion models have a more stable training objective (modeled as maximizing likelihood via denoising) and do not suffer from mode collapse or discriminator oscillation [13]. GAN-based 3T→7T methods often produce sharp images, but they can hallucinate structures or drop details because the adversarial loss is not very sensitive to topological correctness – it only cares if the image *looks plausible*. Our diffusion model, by contrast, directly learns the conditional distribution of 7T given 3T and can be guided to put probability mass on anatomically correct solutions. Recent evidence supports this: the diffusion approach for 3T SWI synthesis achieved image quality on par with real 7T (no significant difference in key vessel visibility) whereas GAN methods struggled with microvascular details [25] [13]. In essence, diffusion gives us high fidelity *and* reliability. By adding our topology-guidance (which GAN papers have not done for this problem), we further ensure that the synthetic images don't just pass visual muster, but also preserve connectivity of structures (no missing gyri or split hippocampi).

- **Why It Beats a Naïve Diffusion (baseline diffusion without special design):** We anticipate that a naive diffusion (just conditioning on the 3T image and training with pixel loss implicitly through diffusion's loss) would improve perceptual quality but might still not guarantee clinical fidelity. Our architectural twist is the injection of **anatomical context**. This could be via adding a segmentation map or edges as additional conditioning channels, or using a hybrid loss (diffusion + deterministic loss). By explicitly including a penalty for topological errors or segmentation differences, we guide the diffusion model to sample images that are not only likely in the pixel sense but also *correct in the anatomical sense*. This addresses a known limitation of vanilla diffusion: it optimizes pixel-level likelihood and may still introduce subtle shape distortions if not constrained. The architecture might also use **latent diffusion** (operating on a lower-dimensional latent space of an autoencoder) to manage memory for 3D data, which speeds up sampling without losing detail – another improvement over a naive high-dimensional diffusion. Overall, our conditional diffusion is **MRI-tailored**: it incorporates domain knowledge (like a segmentation network or known tissue priors) so it outperforms a generic diffusion that doesn't "know" what the structures are.

- **Suitability for MRI and Topology:** Diffusion models by design sample through many small denoising steps, which gives the opportunity to enforce constraints at various steps (e.g. ensure intermediate samples don't stray too far anatomically). They can naturally handle output at the same resolution as input (we can generate full 3D volumes). By using a 3D UNet with attention, the model can capture the global brain layout – ensuring, for instance, if a certain sulcus is present in the 3T image, the model consistently enhances it rather than produces a disjoint pattern. The topology preservation can be enforced by, say, adding a **persistent homology loss** or comparing the Euler characteristic of structures in output vs input (or ground truth) images. Because diffusion models allow easy *guidance* (one can guide generation with an auxiliary model or condition), we can plug in a “segmentation guidance” at generation time if needed. This is harder to do in GANs. In summary, a diffusion framework is very **flexible and controllable**, which is ideal for meeting the strict topology and clinical criteria of our task.

- **Modern Elements Covered:** This direction inherently uses a **foundation-model-like approach** (diffusion is a type of foundation generative model). We can incorporate **Transformer** attention in the UNet. We leverage **multi-modal input** if we condition on T1 and T2. And we can integrate **anatomy constraints** (persistent homology or segmentation nets) easily thanks to the diffusion model’s iterative nature which can accommodate guided sampling. Thus, candidate A aligns with all genuinely modern directions identified earlier.

Candidate B: Multi-Task Transformer-Enhanced GAN (Topology-Aware GAN)

(We include a GAN-based option for comparison, but ultimately will recommend Candidate A over this.)

- **Core Idea:** Develop a **multi-task generative model** that simultaneously produces a synthetic 7T image and a set of auxiliary outputs (such as segmentation probability maps, or even a predicted clinical score), using a shared latent representation. The backbone could be a **Transformer-UNet hybrid**: for example, an encoder that is convolutional to extract local features, plus a Vision Transformer layer to encode global context, and a decoder that outputs the image. We would add a second decoder head that outputs a brain tissue segmentation or other structural representation. The training would use a weighted combination of losses: an adversarial loss (GAN) and reconstruction loss on the image output, plus a segmentation loss (cross-entropy) on the segmentation head, plus possibly a **topology regularization** (for instance, a loss ensuring the segmentation of the synthetic image matches the segmentation of the input or ground truth). Essentially, this is like a **TopoGAN++**: a GAN that is informed by topological features [27] and also by an auxiliary task. We could use a *PatchGAN* discriminator focusing on local texture realism, while the generator is constrained to keep global structure via the multi-task setup.

- **Why It Could Beat Traditional GANs:** By infusing the network with a Transformer component, we allow it to capture relationships across the whole volume (e.g., the symmetry between left and right hippocampi, or the continuity of a vessel) better than a pure CNN GAN. Standard GAN generators can sometimes introduce inconsistencies because convolution is local; a transformer mitigates that. Moreover, the **segmentation multi-task** forces the generator to learn features that are relevant for anatomical structure. It can’t just produce any plausible 7T-looking patch; it must produce one that the segmentation head can decipher correctly. This acts as a regularizer that keeps the generator “honest” to the underlying anatomy. In effect, the segmentation loss is a proxy for topology – if the segmentation of output is wrong (say it predicts two disconnected pieces of hippocampus), that incurs a loss and thus the generator must avoid such outputs. This approach acknowledges the positive results from TopoGAN [27], which demonstrated improved structural fidelity by adding a topological loss, but we integrate it into a modern architecture with transformers and multi-task learning. The GAN’s adversarial part will ensure high-frequency details and realistic textures (sharp edges, proper noise characteristics) that pure L1/L2 losses would smooth out.

- **Why We Still Lean Toward Diffusion over This GAN:** While this Transformer-GAN with multi-task constraint is a strong approach, GAN training is inherently fickle. Ensuring stability while juggling multiple losses (adversarial, reconstruction, segmentation) is complex. Diffusion offers a more straightforward optimization (no min-max game) and has recently surpassed GANs in fidelity. The GAN approach might achieve similar end results, but at the cost of more training tricks (e.g. learning rate tuning, careful loss weighting, possibly needing a perceptual loss to assist, etc.). Additionally, the **diffusion model naturally provides a likelihood-based framework**, which makes quantitative evaluation easier (one could compute log-likelihood or uncertainty in outputs), whereas GANs only offer the discriminator's opinion. Given unlimited compute and time, one could certainly refine a GAN to work, but from a *research novelty perspective*, diffusion is seen as more groundbreaking in 2025. Many reviewers might actually view a GAN, even a fancy one, as a bit behind the times unless it does something truly new.

- **Still, Why Consider Candidate B:** We consider it to make sure we're not dismissing potentially effective solutions. A transformer-based GAN with topology loss would still be novel (no prior work has exactly that for MRI). It directly addresses topology by design and could leverage existing GAN knowledge. If for some reason diffusion proves impractical for 3D (e.g. too slow or data-hungry), this is our plan B. It retains **multi-modal capability** (we can feed T1 and T2 as channels into the generator), and the transformer ensures global context. Importantly, it might be easier to incorporate a **physics-informed block** here: for instance, a differentiable module that mimics a 7T bias field could be inserted in the generator. In diffusion, doing that would be harder as it's iterative. So Candidate B could allow a *modular architecture* where one part explicitly models known MRI physics (like intensity nonlinearity at 7T).

Final Recommendation: We will pursue **Candidate A: the Anatomy-Guided Conditional Diffusion Model** as our primary architecture. This choice is driven by the latest evidence of diffusion models' superior fidelity for high-field MRI synthesis [13], their training stability, and their flexibility in incorporating guidance (segmentation/topology constraints). Candidate A aligns best with a Q1-worthy contribution because it adopts a cutting-edge generative paradigm *and* pushes it further with domain-specific innovation (multi-modal conditioning and topology-aware training). Candidate B (Transformer-based GAN) is a viable alternative, but it carries higher risk (GAN instabilities) and is less aligned with the current momentum in the field. We will keep the insights from Candidate B (e.g. multi-task learning, possible use of transformer blocks) and potentially integrate those into the diffusion framework (for example, using a transformer in the diffusion UNet, or adding an auxiliary segmentation loss to the diffusion training). This way, we harness the best of both worlds while sticking to the fundamentally stronger diffusion approach. In summary, our architecture will be: **a conditional diffusion model with a 3D Transformer-UNet backbone, leveraging multi-contrast inputs and guided by a topology/anatomy preserving loss**.

4. System Topology (Paper-Figure Level Description)

Let's describe the entire system as one would in a figure for a paper, enumerating the modules, data flow, and where key constraints apply. The system can be viewed as a pipeline with distinct stages:

Input & Preprocessing Modules: The pipeline begins with the **3T MRI scans** (for each subject, T1-weighted 3T is the primary input). If we include multi-contrast, the 3T T2-weighted image of the same subject is another input channel. The first module is a *Preprocessing block* (details in Section 5) that takes the raw images and outputs normalized, registered brain volumes (for instance, `I_{3T,T1}` and optionally `I_{3T,T2}`), all aligned in the same space. Alongside the image, we also feed in any available prior information: for example, a **brain mask** or **segmentation map** of the 3T image can be included (the segmentation could be precomputed once using an existing tool on the 3T image). In our dataset, we have

annotation maps (FreeSurfer segmentations) for each subject [34]; these can be used here to provide the model with explicit anatomical labels (though this is optional, it's an added input that could help preserve topology if used as conditioning).

So, the system inputs are: *3T T1w image (aligned, skull-stripped)*, *3T T2w image (aligned)*, and *anatomical priors (mask or segmentation)*. In a figure, this would be the leftmost part, perhaps with two arrows (for T1 and T2) merging into the main network, and a dotted arrow for an optional segmentation prior.

Generative Model (Core Network): Next is the **conditional generation module**. This is depicted as a neural network with an encoder–decoder structure (like a UNet). The 3T images enter the encoder, which progressively downsamples and extracts features. If we use a diffusion model, conceptually we would illustrate it as a series of refinement steps: e.g., a noise input that gets transformed into an image conditioned on the encoder features of the 3T image. However, in a static figure, it might be easier to represent the *denoising UNet at a single step* and note that it's repeated in a diffusion process. Alternatively, we might illustrate the diffusion process as a separate plate: showing that during training, the model takes a noised version of the target 7T and learns to denoise it with the 3T image as context.

For clarity, we can present the core network as a **two-branch UNet**: - One branch encodes the input 3T (and possibly segmentation) into a latent representation capturing multi-scale anatomical features. - The decoder branch outputs the synthetic **7T T1-weighted image** (`I_{7T,T1}^{synth}`).

If multi-task learning is implemented, there could be a **secondary output head** in the decoder that produces, for example, a **segmentation of the synthetic image**. In a figure, this would look like the decoder splitting into two heads: one goes to the synthetic image, another goes to a segmentation map. The segmentation head would be supervised to match a reference segmentation (which we have from the real 7T or from the 3T, since they should be the same anatomical labels).

Within the network, we highlight special components: - **Attention/Transformer module:** at the bottleneck of the UNet (and/or in skip connections), an attention mechanism ensures global coherence. We would label this in the figure as "Global Context Module" that takes all features and reweights them based on global interactions (to ensure, e.g., distributed regions like ventricles or cortex are treated consistently). - **Topology-Preserving Constraint:** We can illustrate this as an auxiliary block during training that checks the output's structures. For example, in the figure, after the synthetic image is output, we apply a "Topology Check" or "Anatomy Loss" by comparing the synthetic image (or its segmentation) to the ground truth. This might be drawn as a loop or a dashed arrow from the output going into a loss function that involves a segmentation network or a known template. In text near the figure, we'd mention: "*Topology loss computed by comparing output segmentation to ground truth segmentation*" or "*persistent homology loss between synthetic and real image*".

Data Flow – Training Path: During training, the **3T input flows through the network to produce the synthetic 7T**. In parallel, the corresponding *ground-truth 7T image* (from the paired dataset) is fed in to compute losses. Specifically: - We calculate an L1/L2 loss directly between `I_{7T}^{synth}` and the real `I_{7T}` (this ensures voxel-wise fidelity for large structures). - The synthetic and real 7T images both can be passed through an external **segmentation model** (say, FreeSurfer or a simpler CNN we train) – we then compute a difference between the segmentations (ensuring that the synthetic image yields the same anatomy labeling as the real). This enforces the topology/structure. In the figure, this could be drawn as both real and synthetic outputs going into a common "Segmentation Module" and then a loss between the two segmentation outputs. - If using adversarial training (less likely in diffusion, but for architecture completeness): the synthetic image would also go to a **Discriminator** module, which compares it to real 7T in terms of realism. The figure might show the discriminator receiving real and fake 7T and outputting a probability, with a GAN loss backpropagated to the generator. - If diffusion, we might illustrate one training iteration where the model tries to predict the noise added to the real 7T image (diffusion target). But that might complicate the figure; we can keep it conceptual: “Our model is trained with a diffusion probabilistic loss and additional supervised losses ensuring image and segmentation consistency.”

Inference Path: At test time (inference), only the 3T inputs are available. The figure should depict that pipeline: the 3T image goes through the generator network, producing the synthetic 7T image. Any auxiliary outputs (like segmentation) can also be produced if we keep those heads active during inference (for example, we might output a predicted segmentation of the synthetic image as a byproduct, which could be useful for analysis). The final output is a **3D volume that mimics a 7T T1 MRI** for that subject. It will have higher resolution/contrast and is aligned to the input.

We should also note **where clinical relevance is injected**: This occurs in two places: 1. **During training via multi-task loss:** The inclusion of segmentation (or a task like Alzheimer’s prediction if we had labels) in the training objective directly ties the image generation to clinical endpoints. In the figure, this was the second decoder head and the segmentation comparison – that is effectively injecting clinical knowledge (since segmentation of hippocampus, etc., is clinically relevant for AD). 2. **During evaluation (and possibly inference):** After obtaining the synthetic 7T, we will perform downstream analyses – e.g., feed it into an Alzheimer’s classifier or morphometry pipeline. While this is outside the core generation model, the system diagram might have a block to represent this in the *evaluation pipeline*. For instance, after synthetic 7T is output, an arrow might go to an “AD Analysis Module” which could be something like “run hippocampal volume measurement or an ML model to predict diagnosis”. We won’t have that inside the training loop (since we don’t have AD labels for training), but it is part of the overall project flow to validate that the synthetic images are useful. So perhaps in a figure, this is drawn as a post-processing: synthetic 7T goes to “Downstream Task (e.g., cortical thickness, AD classifier)” and we show that the performance is improved compared to using 3T image alone.

Where Topology is Enforced: We explicitly enforce topology in the **loss functions** during training. Graphically, one can show a *Topology Loss* coming from either a comparison of segmentations or from a computed topological signature. For example, imagine a “Compute Persistent Homology” block that takes the synthetic and real images and outputs a difference measure that the loss optimizes (as TopoGAN did using MMD between persistence diagrams [35] [27]). Another simpler depiction: highlight that the segmentation loss inherently forces topology preservation. Perhaps annotate the segmentation loss as “(ensures structural consistency)” in the figure legend.

Where Clinical Relevance is Injected: Aside from training-time segmentation guidance, we might incorporate **anatomical indices in the loss**. For example, we could calculate hippocampal volume from the synthetic image (via the segmentation) and compare it to hippocampal volume from the input or known values – penalizing if they differ significantly. This would specifically push the network to preserve the size/shape of the hippocampus, which is relevant to AD. While we might not implement a volume loss explicitly (that could be unstable), this indicates the mindset: every design choice ensures the synthetic image isn’t just visually pretty but quantitatively faithful to clinical markers. In a system diagram, this might not appear as a separate module, but in the text we will emphasize it. The segmentation path covers a lot of it, as volumes can be derived from segmentation.

Data Flow Summary: - **Training:** (3T T1, 3T T2) --> [Generator UNet] --> Synthetic 7T; also (Real 7T) provided for loss. Compute image diff loss, plus segmentation loss: (Synthetic 7T -> SegNet, Real 7T -> SegNet, compare segmentations) for topology; plus adversarial loss if applicable. The figure might show multiple loss arrows coming out of the outputs. - **Inference:** (3T inputs) --> [Generator] --> Synthetic 7T (then used for analysis).

We can describe the system in a figure caption-like paragraph: *"Our topology-preserving 3T-to-7T enhancement framework consists of a conditional generation network and auxiliary loss modules. The 3T MRI (T1w ± T2w) is input into a Transformer-UNet generator, which produces a synthetic 7T T1w image. During training, the synthetic output is supervised with the ground-truth 7T image using intensity-based losses (e.g. L1) and a topology-aware loss. The topology loss is computed by feeding both synthetic and real 7T images into a pre-trained segmentation model and comparing the resulting anatomical label maps, ensuring the synthetic image preserves all brain structures correctly. In inference, only the 3T image is needed to generate a 7T-like image, which can then be fed into downstream analysis pipelines (such as segmentation or Alzheimer's disease classifiers) to evaluate its clinical utility."*

By describing the system at this level, a reader (or a viewer of the conceptual figure) can see all the key pieces: **Input -> Preprocess -> Generator (with possibly multi-task output) -> Synthetic Output** with losses for pixel, topology, and (if used) adversarial, plus an indication that the output goes on to analysis. This modular view will correspond to sections in the paper (preprocessing, model, losses, evaluation methods), demonstrating a clear logical flow from raw data to results.

5. Preprocessing Pipeline (Extremely Detailed)

A robust preprocessing pipeline is critical – not only to feed the model quality data, but also to satisfy reviewers that our comparisons are fair and not confounded by trivial misalignments or artifacts. We will implement **each step with justification** and mention reviewer concerns if skipped. Below is the **step-by-step preprocessing procedure**, aligned with BIDS standards and common neuroimaging best practices:

5.1 BIDS Organization & Field-Strength Identification: Our dataset is organized in BIDS format [36], meaning each subject has sessions "ses-1" (3T) and "ses-2" (7T), and modalities "T1w" and "T2w". We will write preprocessing code to automatically iterate through subjects and sessions in the BIDS directory. The first task is to confirm which session is 3T vs 7T – typically, this is by convention (the dataset authors stated 3T = ses-1, 7T = ses-2 [37]) and can be double-checked via the JSON metadata for each NIfTI (the JSON should contain `"MagneticFieldStrength": 3` or `7`). We'll parse those JSON files to be certain we label data correctly. **Why?** Mis-labeling a 7T as input and 3T as output would invert the problem, so we must be absolutely sure of identifying field strength from metadata.

5.2 Loading & DICOM Conversion: The data providers already converted DICOM to NIfTI using MRIcroGL [38], so we will use their NIfTI files. No need for us to handle DICOM directly (which simplifies things and avoids introducing orientation errors). However, if we had to ingest new raw data, we'd follow similar steps: use dcm2niix or MRIcroGL for DICOM->NIfTI to ensure no scanner-scale bias remains, and preserve orientation info.

5.3 Defacing (Anonymization): The dataset was already defaced with PyDeface [38] (removing facial features). We mention this mainly to show awareness, though it doesn't impact our model training except that some skull parts are removed. It's already done, so no action needed from us, but a reviewer might appreciate knowing privacy was handled.

5.4 Registration & Alignment: *This is crucial.* The dataset provides "Original" and "Aligned" images [34]. The aligned images were produced by linear registration: specifically, they registered 3T T1 and 7T T2 to the 7T T1 space, and then aligned 3T T2 to 3T T1 [38]. In practical terms, all aligned images for a subject share the same voxel space (same orientation, resolution ~0.65 mm, dimensions) – meaning a given voxel index corresponds to the same anatomical point across 3T and 7T images. **We will use these aligned images for training.** This choice is because having input and target pre-registered greatly simplifies learning; the network can do a direct voxel-wise mapping without worrying about shifting structures. If we used unaligned "Original" images, differences between 3T and 7T scans (due to positioning or slight scaling) could be wrongly learned by the model as if they were field-strength differences, which is clearly undesirable. By using the authors' FLIRT-based aligned data, we avoid that pitfall. We will verify the quality of alignment: for each subject, we'll load 3T and 7T images and perhaps subtract or overlay them to ensure major structures line up. The authors mention using FSL FLIRT (6 dof for within-subject presumably) [38], which usually is accurate for same-subject rigid alignment. We anticipate small differences might remain (due to susceptibility distortions at 7T), but they should be minor. If during QC we find any misalignment (say edges of brain don't overlap well), we'll consider a refinement: possibly a nonlinear registration (using ANTs or SPM) of 3T to 7T. However, nonlinearly warping one to another could actually remove real differences (e.g. 7T might have slightly expanded brain edges due to less distortion). We'll likely stick to the provided linear alignment for fairness – since the evaluation should be on the model, not on doing heavy warping.

Justification: If we skip registration (i.e., train on original images), the model would have to learn translation/rotation along with enhancement, which is a waste of capacity and prone to errors. Reviewers would immediately ask: *are you sure differences aren't due to mis-registration?* By using aligned pairs, we preempt that criticism. The Sci Data paper explicitly provides aligned data to facilitate exactly our use-case [34], so not using them would be a mistake.

5.5 Skull-Stripping (Brain Extraction): We will perform skull-stripping on the 3T and 7T images (likely on the aligned 3T T1, and apply the mask to 3T T2 and 7T T1 as well). This yields images where non-brain tissues (skull, scalp) are removed or zeroed out. *Why?* Because the presence of skull and background can confound intensity distributions and network focus. At 7T, fat and bone can appear differently (due to B0 inhomogeneities causing shading or different intensity scaling), and we do not want the network to waste effort modeling how skull brightness changes between 3T and 7T. By stripping the brain, we ensure the model concentrates on brain tissue only – which is where our topology and clinical markers are. Moreover, skull edges might introduce artificial topology (e.g., openings at the foramen or eye sockets) that have nothing to do with brain anatomy. Removing them simplifies any topological loss to just brain structures. We will use a proven tool for this: likely **HD-BET (a deep learning brain extraction)** or FSL's BET on the T1 images. We do have FreeSurfer outputs, which include a brain mask and surfaces [34], so actually we can directly take the FreeSurfer brainmask from the "annotation" data. Using FreeSurfer's mask is robust (FreeSurfer tends to get brain outline right after bias correction). If available, that's ideal to avoid any extra errors.

Reviewers' likely concern: If we *didn't* strip skull, they might worry the model could simply be learning to map the bright skull differently (since 7T might have different coil shading making the skull brighter or darker). That would be a trivial cue ("cheating") that doesn't reflect improving brain detail. Also, including the skull means more heterogeneity in intensities (air vs bone vs tissue) which can destabilize training. One might argue the PLOS 2025 study achieved good results without skull stripping [28] , but they compensated by using segmentation-based evaluation to catch errors. Given our focus on topology, having the skull in images would complicate computing topological similarity (lots of irrelevant connected components from skull). So we will skull-strip to make the problem well-contained. We will, however, mention this decision in the paper and justify it – as leaving the skull could allow the model to produce a full head 7T image. But since our end goal is brain analysis, we are justified in focusing on the brain. If a reviewer asks "does your method rely on skull stripping?", we will respond that it is a standard practice to focus the enhancement on the brain region and that the method in principle could work with whole-head images but would need more data to avoid distractions.

5.6 Bias Field Correction: 7T MRI often suffers from **B1 inhomogeneity**, leading to intensity non-uniformity (parts of the brain appear darker or brighter not due to anatomy but due to coil sensitivity). The dataset used an MP2RAGE sequence for T1 at 7T, which actually mitigates some bias (MP2RAGE provides a uniform image to a degree), but there could still be residual bias. 3T images can have bias too, though less severe. We will apply **N4 bias field correction** (a widely used algorithm) to each image (3T and 7T) after skull-stripping. This will remove smooth intensity gradients. **Why?** We do not want the network to waste capacity learning bias field differences between 3T and 7T. The risk is the network might focus on global shading differences rather than genuine tissue contrast differences. By doing N4, we make the intensities more homogeneous across space, so the model can concentrate on tissue-specific changes. We'll run N4 with standard parameters on each volume. We have to be careful to not overcorrect (N4 can sometimes over-smooth intensities), so we'll verify that after N4, white matter is still distinguishable from gray matter (N4 should preserve contrast, just remove low-frequency bias). If skipping this: reviewers might say "7T has strong bias – did you account for it or is your model output just adding a bias field?" By doing bias correction, we ensure that any intensity differences the model learns are presumably due to tissue contrast, not an artifact of coil profiles.

5.7 Intensity Normalization: This step standardizes the range and distribution of intensities across subjects and modalities. We plan a two-pronged approach: - **Within-image normalization:** For each image, we will scale/normalize intensities to a consistent scale. A robust method is to compute the mean and standard deviation of intensities within the brain mask and then subtract the mean and divide by std (z-score normalization). Alternatively, we could normalize by a percentile (e.g., set the 99th percentile intensity to 1.0, and 1st percentile to 0) to roughly map each image's dynamic range to [0,1]. We need to be cautious: 7T images might have different overall contrast, but if we independently scale each image, we might remove that difference. We likely choose a scheme that preserves relative contrast differences between 3T and 7T while removing subject-specific variability. One approach: use the 3T image's brain mean as a reference – scale all 3T images to mean 0, std 1, and apply the same linear scaling to the 7T of that subject (since 3T and 7T are paired, we could normalize them together). However, since the physical units are arbitrary, simpler is: perform a standard intensity normalization separately for input and output domains. We will do: *for 3T images, subtract mean and divide by std; likewise for 7T images*. This yields both in a zero-mean, unit-variance space. The network training (especially diffusion) benefits from inputs roughly in a similar numeric range. We will, of course, apply the same normalization consistently to training and validation data (computed per image to avoid information leakage across subjects). - **Cross-subject normalization:** After z-scoring, we might still want to ensure no outlier intensities (maybe due to some noise spikes) throw things off. We can clamp extreme values (e.g. anything beyond ±5 sigma can be clipped) for stability. Given MRI intensities roughly follow a tissue distribution, this just prevents occasional weird voxels from affecting training.

Rationale: If we skip normalization, the network has to handle potentially large intensity scale differences between subjects or modalities. One subject's 7T might be overall brighter if scanner gain was different, etc. That can slow training or cause one batch to dominate loss. Normalizing addresses a reviewer concern: *"Did you intensity-normalize? Because otherwise your results might reflect trivial intensity scaling."* We will explicitly say we did, so reviewers know our model isn't just learning to multiply intensities by a factor (which would not be publishable). Instead, it has to truly alter the image's detailed contrast and texture.

We will note that we do preserve the fundamental *contrast difference* between 3T and 7T in a relative sense. For example, at 7T gray matter vs white matter intensity ratio may be higher than at 3T [15]. By normalizing each image to its own stats, do we lose that? Actually, if we z-score each image, we remove absolute differences in contrast amplitude. Another strategy: perform **histogram matching** from 3T to 7T globally, but that could actually remove differences we want the model to learn. Better is to let the model learn the contrast change. So perhaps a compromise: we do a mild normalization (like each image median to 1) but not fully equalizing contrasts. We might do: *normalize each image by dividing by its median brain intensity*. This will put all images on roughly similar scale (since median of brain might correlate across subjects), but it won't force the distributions to unit variance. Then the network can learn contrast differences. For clarity to reviewers, we'll mention we tried a few normalization schemes and picked one that gave best results on validation (we'll justify selection in the paper). The key is we **explain our choice**. Likely, saying "we applied z-score normalization of each volume's intensities within the brain mask" is acceptable, as many works do that. We'll then rely on the network to handle contrast differences via training on paired data.

5.8 Resolution Harmonization: The 3T and 7T images originally had different resolution (0.8 mm vs 0.65 mm). In the aligned images, the authors resampled everything to 0.65 mm isotropic (the 7T T1 reference space) [39] [38]. Therefore, *our data is already resolution-matched* – every aligned image has the same voxel size and dimensions. This is perfect for our model, since we want to output an image at the higher 7T resolution. We will double-check one detail: when upsampling the 3T 0.8→0.65 mm, interpolation was used (likely trilinear in FLIRT). This means the 3T images have been slightly super-sampled. They will still look smoother than the true 7T because they contain no new high-frequency info, but at least they are on the same grid. We will not alter the resolution further – we will train the model to produce outputs on that 0.65 mm grid. If some reviewers are very pedantic, they might ask: "So you effectively are doing super-resolution from 0.8 to 0.65 mm? Are you sure the difference is due to field strength, not just resolution?" We can answer: 0.8 vs 0.65 is a minor difference in linear dimension (~20% finer). The field strength also affects contrast and SNR, which our model accounts for. And importantly, our input is already resampled to the output resolution, so the model's task is not simply interpolation – it must generate details that were missing. We could mention that point. No further smoothing or anything is needed; we trust the provided alignment pipeline (which used sinc interpolation likely by default).

If we skipped this harmonization (i.e., left 3T at 0.8 and 7T at 0.65), the model would have had to do both *upsampling and enhancement*. We decoupled the upsampling by doing it via registration. This is a legitimate approach and we'll clarify it in the methods.

5.9 Optional Spatial Normalization: This is about whether to align images to a common template (like MNI space). The dataset is in each subject's native space after alignment (subject space). We likely keep it that way to preserve subject-specific fine details without additional interpolation. We do not need to bring to MNI, since we have one-to-one pairs. Some pipelines might warp to MNI for ease of evaluating metrics across subjects (voxel-wise), but that would distort individual topology slightly. We prefer to operate in native space. So we will *not* do a nonlinear normalization to a template. Instead, we evaluate metrics either in native space (which is fine since images are paired) or by using the provided surfaces/segmentations for each subject. This avoids any unnecessary degradation. We note this decision in case reviewers wonder. We justify: given only 10 subjects, adding a group-wise registration step could reduce the already limited unique details (makes brains more similar, possibly hiding differences). We want to preserve genuine anatomical variability for the model to handle.

5.10 Quality Control & Logging: Throughout preprocessing, we will implement thorough QC: - After skull stripping, we will generate quicklook images (like axial slices) with the brain mask overlay to ensure no brain tissue was cut off and minimal non-brain remains. We'll log if any subject had poor mask (we can manually fix that subject by tweaking the mask or using the FreeSurfer mask). - After registration, we can create an overlay: e.g., overlap 3T and 7T edges or use a checkerboard visualization to verify alignment. We expect very close alignment given rigid reg on same subject; any large mismatch we flag. If a subject shows mismatch, we might rerun FLIRT or use a different cost function (maybe 7T T1 vs 3T T1 might have contrast differences that confused FLIRT; if that happened, we can actually align via the T2 modalities or use mutual information). The Sci Data authors likely did a good job, but we won't assume blindly. - We will record basic image stats (mean, std, SNR from MRIQC JSON) before and after normalization to ensure the normalization did what we expect (for instance, after z-scoring, means should be ~0). - We'll also inspect intensity histograms of 3T vs 7T after preprocessing. Ideally, the 7T should show higher gray-white contrast. If our normalization accidentally removed that (e.g., if both end up similar histograms), we might adjust the strategy. For example, if z-scoring each image, both 3T and 7T will have mean0, std1, so their histograms are standardized – the difference will then be subtle (maybe higher-order moments differ). Maybe that's okay as the network can still learn that GM intensities relative to WM differ. We just need to ensure we haven't made 3T and 7T trivially identical in distribution. A possible tweak: only normalize 3T, and apply same linear scaling to 7T, preserving ratio. But since pair sizes are small, per-image normalization is fine.

- We'll keep an **error log**: if any step fails (e.g. one subject's FreeSurfer segmentation might have failed – though authors likely ensured it succeeded for all), we log and address it. If something like MRIQC flagged a poor image (maybe motion or artifact), we note that. All our data are healthy adults, so likely high quality, but one never knows (e.g., some 7T could have lower SNR in temporal lobes due to B1 drop-off).

5.11 Preprocessing Fail-Safes: We anticipate possible failure modes: - **Skull-strip failure:** If the algorithm mistakenly erodes brain (e.g. sometimes frontal poles get cut if contrast is weird), the synthetic image training could suffer because input brain is incomplete. To detect this, we'll specifically check the brain mask volume or edges. We can use the T2 image or combined probability to improve mask if needed (since T2 might help find CSF vs skull). In case of any doubt, we prefer using the FreeSurfer-derived mask which is usually reliable. - **Registration error:** If one subject's 7T had distortion that linear reg couldn't fix, we might see a slight misalignment in outer cortex. To detect, we rely on QC overlays. If misalignment is small, it might not kill the model, but we strive for pixel-accurate alignment. We could do an optional nonlinear warp from 3T to 7T T1 for that subject. But note: Nonlinear warp would basically give the model an even easier task (as it would correct for any shape differences too). That might be too much – arguably, we *want* the model to possibly learn any systematic shape difference due to, say, susceptibility (e.g., maybe 7T brains appear slightly larger in certain regions or ventricles shape a bit differently?). However, those differences are probably negligible between 3T and 7T for structural images, if both are high-res. - **Intensity outlier:** If any image has unusual intensity scaling (maybe one subject's 7T was acquired differently), MRIQC metrics would catch it (like high CJV or EFC [40]). We will examine MRIQC outputs (they provided JSON with IQMs [41]). If, say, one subject's 7T had a very high noise, we might consider excluding that subject from training to not confuse the model, or handle it separately. But with only 10 subjects, we'd try to include all unless truly problematic.

5.12 Reviewer Hotspots in Preprocessing: We explicitly note in our write-up that we addressed common criticisms: - *"Did you ensure 3T and 7T are aligned?"* – Yes, using provided aligned images (rigid registration) [38] . - *"Did you remove non-brain elements that could bias results?"* – Yes, skull stripped and bias-corrected to focus on brain tissue. - *"Intensity normalization?"* – Yes, we normalized intensities to a common scale, avoiding trivial intensity mapping. - *"Reproducibility of preprocessing?"* – We will mention tool versions (FSL, ANTs, etc.). For example, we'll state we used FSL FLIRT with 6 DOF and correlation ratio cost, N4 from ANTs with default 50 iterations, etc., so others can replicate. Also mention any manual interventions if done. - *"What happens if a preprocessing step is skipped?"* – We can preempt this by short reasoning: if no skull strip, how would it harm? If no alignment, how? (We might even run an experiment: train a model without alignment and show it fails, but that might not be necessary to include in paper, can just reason). - *"Are the images pre-filtered or denoised?"* – We haven't mentioned denoising; we could consider applying a mild denoising if needed. However, the images likely are already pretty clean (MP2RAGE is quite clean). Too much filtering could remove detail. We likely skip any extra denoising, except what bias correction does. We will however mention that no additional smoothing was applied (so we don't inadvertently remove fine details that the model should learn to create). - *"How do you ensure quality consistency across the dataset?"* – We rely on the authors' MRIQC. They reported that all images have good contrasts and structural detail [40] . We'll cite that all images had acceptable MRIQC metrics, meaning no major artifacts.

In conclusion, our preprocessing pipeline takes raw paired 3T/7T data and produces **clean, aligned, normalized brain volumes** ready for model training. Each step (defacing, alignment, skull stripping, bias correction, normalization) exists for a reason: to remove confounds and reduce the problem to its essence (field-strength related differences). We won't allow sloppy preprocessing to undermine the science – by the time we feed the data to the network, we have high confidence that any differences to be learned are *true differences* due to 3T vs 7T, not due to position, orientation, or other artifacts [38] [34] . This meticulous approach will reassure reviewers that our results are attributable to the model's capabilities, not preprocessing quirks.

6. Data Formulation for Learning

With preprocessed data in hand, we need to define how we will present it to the model during training, ensuring no information leakage and making the most of our limited sample size. Key considerations are how to pair inputs and targets, whether to use patches or full volumes, and how to incorporate multi-contrast information.

6.1 Input–Target Construction: Each training sample will consist of an input (source domain) and a target (ground truth in target domain). For our main formulation: - **Input = 3T T1-weighted MRI** (after preprocessing, aligned & normalized). - **Target = 7T T1-weighted MRI** of the same subject (aligned & normalized identically). These form a pair. If using additional modalities: - We can augment the input by including the **3T T2-weighted MRI** as a second channel. In practice, the network's input tensor could be 2-channel (channel0 = T1_3T, channel1 = T2_3T), and the output is 1-channel (T1_7T). - We could also feed in the **brain mask or segmentation** as extra channels if we choose. For example, a simplistic approach: multiply the input by its mask (effectively done by skull stripping already). Or append a "segmentation" channel where each voxel contains an index or probability of tissue class. However, providing segmentation as input might leak too much high-level information and risk the network overly relying on it. Instead, we prefer to use segmentation in the loss, not necessarily as an input (though one could input a mask to delineate brain boundary clearly).

We will formulate the **training pairs** as strictly within-subject pairs of images. There are 10 subjects × (in principle) maybe 1 image per subject (though each has two sessions: we have effectively 10 pairs if each subject yields one 3T and one 7T). Actually, our dataset has possibly two sessions (maybe repeated scans?), but it says ses-1 and ses-2 correspond to field strengths, not separate time points with same field. So each subject yields exactly 1 paired example. That's extremely small (N=10). To mitigate this, we will exploit the 3D nature of the data: - We can extract multiple **patches or sub-volumes** from each 3D image, treating each patch as a training sample (with corresponding patch in target). - Or we can use data augmentation to simulate more examples (see below).

6.2 Patch vs. Full-Volume Training: Training on full 3D volumes (which are ~300×300×250 voxels) with heavy models like diffusion or transformers is computationally expensive and might not fit in GPU memory, especially with diffusion's multiple time steps. Also, with only 10 volumes, full-volume training might overfit quickly and has only 10 samples per epoch. Instead, we will adopt a **patch-based training strategy**: - We will sample many small 3D sub-volumes (cubes) from each subject's data. For instance, patches of size 64×64×64 or 96×96×96 voxels. Each patch from the 3T forms the input, and the corresponding patch location from the 7T forms the target. - Because the data is aligned, extracting a patch at a certain coordinate from 3T and the same coordinate from 7T yields a perfectly matched pair of regions. - This dramatically increases the number of training samples: if each volume yields thousands of possible patches, we effectively augment our dataset size. - We will ensure patches cover all parts of the brain over training iterations, not just one region. We'll use random sampling for patch centers, possibly biased to pick mostly brain voxels (we don't need patches that are mostly background outside the brain). - Patches allow us to incorporate translation invariance: the network can learn locally and it doesn't matter where in the brain the patch came from, as long as the context in that patch is enough. One caveat: some global context might be needed (e.g., to know if a patch is from frontal lobe vs occipital, if that matters for AD or bias fields). We partly address that by using a network with some global awareness (the transformer or large receptive field). - We might also consider multi-scale patch approach: e.g., 64^3 patches for fine detail learning, and occasional larger patches or downsampled full volumes to capture coarse structure.

Why not train on full images? Primarily memory and data efficiency. Patch training is a common strategy in super-resolution tasks and was used in the literature: the PLOS 2025 study used 64×64×64 patches in a 3D U-Net and found it optimal for low-data regime [42] [43]. By using patches, we also effectively do *data augmentation by position* – each patch can be seen as a shifted version of the volume. We just have to be careful at inference to tile the output (discussed below).

6.3 Multi-Contrast Usage (T1+T2): If we incorporate the T2 channel, each patch will include corresponding T2 voxels. The aligned dataset ensures that for any given patch coordinates, we have T1_3T and T2_3T and T1_7T. We suspect adding T2 could help, because T2 might highlight different tissue boundaries (e.g., CSF vs tissue edges). For instance, if a subtle structure is not well-defined in T1 input, T2 might give a complementary cue, enabling the model to synthesize it more accurately in T1 output. However, we must weigh complexity: requiring T2 means the approach is only directly applicable if both contrasts are available. Many clinical scans do include both T1 and T2, so it's not unreasonable. We will likely explore two modes: - **Single-modal input** (T1→T1) as the primary path (for generality, and because T1 is key for AD). - **Multi-modal input** (T1+T2 → T1) as an experiment to see if it boosts performance. This can be an ablation study: does the T2 help significantly? We anticipate some improvement in visual quality or certain regions (maybe white matter lesions or CSF spaces). - If the improvement is big, we could include multi-modal as a main result. If it's minor, we might keep the method single-modal for simplicity and mention the multi-modal in discussion.

Choice rationalization: We lean toward using **T1w as the main target** because early AD changes (like atrophy) are primarily assessed on T1. T2 could be used to identify vascular lesions or edema, but for AD specifically, T2 is less critical. So our synthetic target is only T1 7T. We're not, for example, trying to also synthesize 7T T2 (though that's theoretically possible). That would double output channels and complexity, with little AD payoff. So **one target (7T T1)** is the plan.

6.4 Avoiding Data Leakage: With such a small dataset, we must be vigilant about train/test separation. We will set aside some subjects as test (and possibly some as validation). A likely scheme: use a **cross-validation** or leave-one-subject-out approach. For example, train on 8 subjects, validate on 1, test on 1, and rotate. Or train on 9, test on 1, and do 10-fold LOOCV given only 10 subjects. For final results, we might report the average over folds. Alternatively, since each subject yields many patches, we could do an 80/20 split at the patch level – *but that is incorrect*, because patches from the same subject would appear in both train and test, causing leakage. So we will do splitting at subject level, not at patch level, to ensure no leakage. Each subject's images are treated as a unit.

Important: We must ensure that if we use any augmentations (like flips), we do them only on training patches, not on validation/test, obviously. Also, if we do any hyperparameter tuning, we use a separate validation set (or cross-val as above) to avoid biasing the test results.

We will also avoid using *7T data in any way to augment 3T in training except paired as target.* The semi-supervised idea from literature was to use unpaired 7T volumes to help. If we had extra 7T data (the PLOS paper used 20 unpaired 7T volumes [44]), we could incorporate that via an adversarial or consistency loss to teach the generator what real 7T distribution looks like beyond our 10. In our case, we might consider using the existing 10 7T themselves as "unpaired" in some semi-supervised fashion to augment. But since they're paired anyway, that wouldn't add new info. If we get access to any other public 7T data (maybe Human Connectome 7T images or something), we could incorporate them as unpaired to strengthen the diffusion model's prior. That's a potential extension if needed. We'll mention domain adaptation possibility, but careful: it could be out of scope to implement fully. For now, our plan is purely supervised with these 10 pairs.

6.5 Handling the Small-N Regime: With only 10 subjects, overfitting is a major risk. We tackle it in multiple ways: - **Data Augmentation:** We will apply aggressive augmentation to the training patches. This can include: - *Spatial transformations:* random small rotations (e.g., ±10 degrees), random flips (LR flip effectively swaps hemispheres – since our model shouldn't assume left-right asymmetry except maybe for pathology, but in healthy brains it's okay), random scaling (a few percent zoom in/out), and elastic deformations (to simulate slight anatomical variability). However, one must be cautious: a large elastic warp could make the 3T–7T pair no longer exactly aligned. Ideally, we apply identical augmentation to both input and target patch, so the mapping remains correct. Yes, we will do paired augmentations: any spatial transform applied to the 3T patch will be applied to the 7T patch in the same way (since we know the ground truth output). - *Intensity augmentations:* random bias field addition (simulate some intensity non-uniformity, again applied to both input and target to not confuse mapping? Or perhaps applied only to input to simulate different 3T coil effects – but that breaks the exact correspondence, so better to apply identically or not at all). We could also add Gaussian noise, though that might not be necessary if images already have noise; still, adding slight noise can make the model robust to noise variations. - *Dropout of modality:* If using T1+T2 input, maybe sometimes drop T2 to force reliance on T1; but that's more advanced, probably not needed. - *Blur or resolution variation:* We could slightly blur the 3T input patch with a random small kernel to simulate variability in scanner resolution. But since our data is consistent and we pre-fixed resolution, we might skip that. - These augmentations effectively create an infinite stream of unique training samples from 10 subjects. We have to ensure we do not do augmentation on validation/test (except maybe the flips if we are doing cross-val and count that separately). - **Semi-supervised or transfer learning:** Another approach from literature is to use unpaired data. If needed, and if we find publicly available 7T images (like from another study), we could incorporate them with a cycle-consistency or adversarial training. For instance, train a diffusion or GAN on unpaired 3T and 7T to capture distribution, then fine-tune on our paired small set. The PLOS paper did something similar by adding 20 unpaired 7T and employing a consistency loss [45] [44]. We will consider this if baseline training on 10 paired is not sufficient. It might complicate training but it's a backup to improve generalization. If we do, we have to ensure any additional data is properly acknowledged and its differences (site, etc.) accounted for. - **Cross-Validation for Evaluation:** Because any single split could be unstable (with 10 subjects, picking 2 for test might drastically swing results depending on which), we likely will do LOOCV or repeated experiments. This not only helps robustness of evaluation but also *uses data more efficiently*: eventually, every subject gets to be in training in some folds. However, one must be careful – if we tune hyperparameters on cross-val, we might still overfit to these 10 in general. But given the scarcity, cross-val is a fair approach to show consistency. Reviewers often worry about small N, but cross-val is an acceptable mitigation if properly done (report mean and std of metrics). - **Regularization in model:** We will use weight decay and maybe dropout in the model's layers to reduce overfitting. Also, the multi-task (segmentation) acts as a regularizer – it prevents the network from just blindly copying input intensities because it has to also make meaningful segmentation. That should help low-data learning by injecting prior knowledge.

6.6 Patch Aggregation at Inference: One must consider how to reconstruct a full 7T volume from patches at test time: - Easiest is to slide a window over the 3T volume, generating patches, run each through the network, and place them into an output volume. If we do that, we need to blend the patch outputs to avoid seam artifacts. Commonly, one would average overlapping regions or use a blending window (like raised cosine) on patch edges. - Alternatively, if we have enough GPU memory at inference, we could process the whole volume in one go by stacking the operations (some frameworks allow large input if done carefully). With diffusion, we might try to do patch-by-patch for sampling as well. - For now, we'll plan to use *patch inference with overlap*. We'll probably pick an overlap of 50% so that each voxel is predicted multiple times from different patches and then average. This smooths any patch boundary differences. We should verify the model is approximately shift-invariant (likely yes if using conv and appropriate handling). - Because the data is small, we can also simply evaluate on a patch basis if needed (like compute SSIM per patch and average). But better to reconstruct the full image and compute metrics on that to account for global consistency.

6.7 Avoiding Information Leakage in Multi-Task: If we use segmentation in training, we must ensure the segmentation labels we use are from the training set only. Since segmentations are derived from the same MRI, it's fine. But consider: if we used a pre-trained segmentation model (like from all subjects) and applied to input, that's not really leakage because segmentation is a deterministic function of input, not using test outputs. So that's okay. But if we were to incorporate something like "we know the hippocampal volume of the target and use that in loss", that would be using target info beyond what input provides. That could be considered as peeking at ground truth labels. However, since we have ground truth 7T during training, that's fine to use in loss. In testing, obviously we don't use ground truth. So no issue as long as all supervised signals are legitimate.

6.8 Small-N Handling with Pretraining: Another thought: We could pretrain the model on a similar task with more data. For example, pretrain on synthesizing T2 from T1 (like GAN-MAT did) on a larger dataset [26], or pretrain on 1.5T to 3T (some bigger dataset might exist for 1.5T to 3T). Or even use a self-supervised pretraining on the 3T images alone (like masked autoencoding or denoising on 3T). Then fine-tune on our 10 paired data for 3T→7T. This could give the model a starting point of general MRI feature extraction. If time permits, we could do something like train a VQ-VAE on a bunch of open 3T MRIs and use it in a latent diffusion. But that might be beyond scope unless needed for performance. We will mention it as a possible extension if reviewers are concerned about small data – showing that we have considered using larger datasets via transfer learning.

6.9 Ensuring the Model Doesn't Cheat: "Cheating" here means the model exploiting any unintended clues. Since we have perfect alignment and same resolution, one might think there's no obvious cheat (like watermark etc.). But subtle: if we left some header info (like DICOM tag saying field strength) – not applicable here since input is just image array. Or if intensity ranges were not normalized, the model could possibly just learn a global scaling to mimic 7T mean intensity (if all 7T have say higher mean). We normalized to stop that. We also ensure that any augmentation doesn't introduce scenario where input contains part of output. Paired augmentation is safe. So likely no cheating paths remain.

6.10 Pairs and Batches: We will treat each patch pair independently in training. For diffusion, each will generate many noise-augmented pairs internally. We'll shuffle patches each epoch. A batch may contain patches from different subjects, which is fine after normalization (they're all roughly standardized). We must ensure batch mixing doesn't cause an issue with losses – it shouldn't, as each patch has its own target.

6.11 Summarizing Data Preparation: We have 10 paired volumes, we generate thousands of paired patches from them for training, using both T1 and optionally T2 as inputs, with each patch's target being the corresponding 7T T1 patch. The small sample size is addressed by heavy patch sampling and augmentation. Data splits are by subject to avoid any overlap. This way, when we evaluate on a left-out subject, the model has never seen any part of that subject's image during training, giving a fair generalization test.

Reviewers' likely questions on data formulation: - *"Why patches? The brain has global context – could patch training miss that?"* We will justify that our network (with transformer attention) still captures large context even from a patch, and our patch size (e.g. 96^3) is large enough to include significant anatomical context (it might cover, say, half a hemisphere). Also, patch training is a proven strategy in similar works [24] [42], especially given GPU constraints. - *"Could overlapping patch predictions cause seams?"* We will note that at inference we use overlap and blending to avoid seams. We can also mention we didn't observe discontinuities in synthetic images (and we can show a full reconstruction in the paper's figures to demonstrate it looks natural). - *"Why not use all data in one network (like 10 images is not huge)?"* If asked, we can say we did try or consider full volume training, but patch training gave better generalization and allowed more effective data augmentation. Also, full 3D diffusion with 10 volumes might severely overfit or be unstable; patches effectively multiply our training examples. - *"Multi-contrast – is it fair to require T2?* – We can respond that it's an extension we explored, but our core results use only T1 as input, meaning our method works with the minimal common clinical acquisition (T1). The T2 input variant is optional for situations where T2 is available and one desires maximum accuracy. - *"Data leakage and evaluation fairness?"*

– We will clearly state that test subjects were held-out and that we perhaps did cross-validation due to sample size. We'll also mention the use of statistical testing to ensure results are not noise (discussed in Evaluation).

By carefully constructing our training data pipeline as above, we ensure the model learns the legitimate 3T→7T mapping without distraction, and that our evaluation is rigorous given the small N scenario.

7. Training Strategy

Training a high-capacity generative model on limited data requires a well-planned strategy. We will outline the loss functions, training phases, and regularization techniques to ensure convergence to a model that meets our objectives (visual fidelity, anatomical accuracy, and clinical relevance). We will also discuss how to detect and address failure patterns during training.

7.1 Loss Functions:

Our training loss will be a weighted combination of several components, each serving a purpose:

- **Pixel-wise Reconstruction Loss:** We will use an L1 (mean absolute error) loss between the synthetic 7T image and the ground-truth 7T image. L1 is chosen over L2 because L1 tends to preserve edges better and is more robust to outliers (L2 would overly penalize occasional spikes, potentially blurring the output). This loss ensures *global and local intensity accuracy*. It trains the model to get tissue intensities and contrasts right on average. We expect L1 to enforce large-scale structures (e.g., brain shape, ventricle size) to match, since any deviation is penalized.

- **Perceptual (Feature) Loss:** A perceptual loss compares high-level features of output and target, rather than raw pixels. We will implement this by passing synthetic and real 7T images through a **pre-trained feature extractor** and computing the difference in feature activations [46]. A common choice is the VGG-19 network pretrained on ImageNet, taking a certain layer's output (e.g., conv3_3) [46]. Even though VGG is trained on natural images, it has been surprisingly effective as a perceptual metric for medical images as well, capturing textural similarity. Alternatively or additionally, we could use a network trained on brain MRI segmentation (like a shallow network we train on our data's labels) as a feature extractor. But to keep it simple, VGG or a standard perceptual loss (LPIPS) can be used. This loss encourages the synthetic image to *feel* like the real image in terms of texture and fine details, beyond just pixel matching. It helps prevent the output from being overly smooth (a typical result of L1/L2 alone). We'll weigh this loss such that it complements L1: too high and it might introduce noise/artifacts chasing perceptual similarity, too low and it has no effect. We might start with a small weight (like 0.1 relative to L1) and tune on validation.

- **Topology/Anatomy Loss:** This is the crux of our **topology-preserving objective**. We propose two implementations and might use one or a combination:

- **Segmentation Consistency Loss:** As discussed, we take a segmentation model (like FastSurfer or a simpler UNet trained to segment brain structures) and generate label maps for both synthetic and real 7T. Let's denote S_{synth} and S_{real} as the segmentation outputs (could be one-hot label volumes or probability maps). We then compute a loss $L_{seg} = \mathcal{L}(S_{synth}, S_{real})$. If we have discrete labels, a straightforward choice is the **Dice loss** or cross-entropy on each class, comparing synthetic's segmentation to the real's segmentation (which we have from the dataset annotations). Essentially, we want them to match. This ensures things like: if the real 7T had a single connected hippocampus segment, the synthetic 7T's intensities must be such that a segmentation model also finds a single connected hippocampus of the same size. This is a direct measure of topology and anatomy. We might simplify by focusing on key regions: e.g., restrict this loss to certain "regions of interest" that matter (like hippocampus, ventricles, cortex), which we can weight more heavily. If using all brain regions, the model is encouraged to get all structures right – which is great for topology, though we must ensure the segmentation of real data is accurate (FreeSurfer can have minor errors, but we assume it's decent).

- **Direct Topological Loss (Persistent Homology):** For a more direct approach, we can calculate topological features of the images. For example, using a persistence diagram of sub-level sets of the image or segmented structures. A simpler measure could be counting connected components or holes in a thresholded version of the image (like binarizing at certain intensity to approximate white matter). However, doing this directly on intensity is tricky because intensity doesn't directly give topology of anatomy unless segmented. So a better approach might be computing topology on the segmentation as above. The literature suggests using persistent homology metrics and maximizing their similarity [47] [27]. We could implement: compute persistence diagrams for cortical gray matter surfaces or something and penalize differences. This is complex and computationally heavy, so a more practical surrogate is the segmentation loss above, which inherently tries to preserve shape connectivity.

Another idea: incorporate a **surface distance loss** using the provided pial/white surfaces [34]. We have the actual cortical surface from real 7T (via FreeSurfer). We could require that if we run the same surface extraction on synthetic, the vertex coordinates align. But running FreeSurfer inside training is not feasible. Instead, maybe sample points from real surfaces and ensure synthetic intensities have appropriate gradients at those points to indicate a surface. That's too convoluted. So we stick to segmentation-based.

We'll implement L_{seg} by perhaps training a small 3D UNet to map T1 to a few label classes (e.g., background, gray, white, ventricle) using the annotation maps. Then hold that network fixed during training and compare outputs. Or we could skip training our own and directly use the FreeSurfer labels as S_{real} and attempt to predict them from synthetic by including a segmentation head in our generator (multi-task as earlier). Actually, the multi-task approach is elegant: *the generator itself outputs a segmentation and an image*. Then we have ground truth labels from real 7T to compare to the segmentation output (with cross-entropy). This way we don't need a separate segmentation network, and the generator is jointly learning to ensure those labels are correct. This effectively couples the image generation with segmentation implicitly – if the image does not align with those labels, the segmentation head will struggle and incur loss, pushing the image to improve. We likely will do this multi-task design as it is simpler to implement and efficient. The **topology loss weight** will be set such that the model doesn't ignore it. Possibly initially lower weight then increased once image starts to be reasonable (we can use a curriculum for that, see below).

To justify, the PLOS study introduced segmentation-based metrics after training to evaluate [28], but did not enforce it in training; we go further by including it in training to directly optimize for that.

- **Adversarial Loss (if applicable):** If we incorporate a GAN discriminator (particularly if we ended up with Candidate B or a hybrid approach), this would be a term L_{GAN} that encourages the synthetic output to be indistinguishable from real 7T. It typically is of the form: for the generator, minimize $-\log D(I_{7T}^{synth})$ (if using a log-loss) or an equivalent WGAN-GP loss etc. This pushes the generator to produce realistic textures that fool D. As we favor diffusion, we might not include a discriminator. However, one could still incorporate an adversarial element even with diffusion by fine-tuning with GAN for sharper details. But let's say our primary plan is *no adversarial loss at first*. If we see outputs are too smooth, we might add a *PatchGAN* discriminator that focuses on high-frequency detail (like edges, fine patterns in cortex). That discriminator would look at small patches of the synthetic vs real to see if textures (like gyral patterns, micro vessels) look real. We need to carefully weight adversarial loss to not conflict with segmentation loss (adversarial might try to invent "realistic" textures that segmentation might not expect). A balanced approach: use a *low-weight adversarial loss* primarily to add variation. If diffusion is used, we might not need this at all; diffusion inherently can produce noise patterns well. We will keep it as an optional component in case.

- **Regularization Losses:** These include weight decay on network weights (L2 norm of weights, typically already included in optimizer settings). Also, if using diffusion, it has its own internal loss (the KL or noise prediction loss) which is primary. Actually, clarify: In diffusion training, the main loss is the MSE between predicted noise and true noise at random timesteps. If we use a conditional diffusion, that's our base loss. So when saying L1/L2 loss above, we must reconcile with diffusion:

- In diffusion model training, we do not explicitly have an L1 image loss; instead, we train the model to denoise. However, to incorporate our additional constraints (segmentation loss, etc.), we can treat those as *auxiliary losses added to the diffusion loss*. This means total loss = diffusion denoising loss + λ_seg * segmentation loss + etc. This is an uncommon but feasible approach: we are basically guiding the diffusion model's training with extra terms.
- Alternatively, we could do a two-stage training: first train the diffusion purely on pixel loss (denoising), then fine-tune it with segmentation constraint. But joint training should be fine if weighted properly.

We also consider **feature matching loss** (used often in GANs): if a discriminator is used, one can add a loss that the generator tries to match intermediate features of the real vs fake (stabilizing training). If needed, we can incorporate that to help the generator produce more realistic outputs without strictly adversarial push.

7.2 Multi-Task Learning: We have effectively a multi-task setup if we output both the synthetic image and the segmentation. The segmentation head introduces an additional supervised signal (with cross-entropy or Dice loss with the ground truth labels from real 7T). This multi-task approach has multiple benefits: - It acts as regularization (network must learn meaningful features for segmentation, not just brute-force pixel mapping). - It ensures anatomical correctness explicitly. - It provides an additional evaluation output for free (we can check if segmentation performance on synthetic vs real are close).

We will design the network such that the encoder is shared, and then we have two decoders: one for image reconstruction and one for segmentation. The segmentation decoder could be a lighter branch (since segmentation is easier in some sense than generating intensities). They may share early decoder layers and only branch out near the end.

We'll weigh the segmentation loss relative to image loss to ensure neither dominates. In early training, the image may be very noisy (especially in diffusion) which will make segmentation difficult; if segmentation loss is too high early, it might confuse training. We might do a **staged training** where segmentation loss is introduced or increased after some epochs (see curriculum below).

7.3 Regularization Strategies: - **Weight Decay:** Use a modest weight decay (e.g., 1e-5 or 1e-6) to keep model weights from growing and to encourage simpler solutions. - **Dropout:** If using dropout (especially if a transformer or any fully connected layers present), we can drop some activations during training to generalize better. For convolutional layers, dropout can still be applied (e.g., drop filters randomly). Perhaps not crucial in diffusion (which has inherent noise injection), but if we go with a GAN or direct network, dropout might help. We'll monitor if overfitting occurs (loss on train >> loss on val). - **Augmentation as reg:** Already discussed heavy augmentation which is a key regularizer given small data. - **Loss weighting:** We consider that effectively as reg – too high weight on adversarial could cause instability, too high on segmentation could cause the network to ignore image fidelity. We'll tune these on validation: e.g., try λ_{seg} = 1.0 vs 0.1, see which yields better val SSIM and segmentation accuracy trade-off. - **Early Stopping:** Given risk of overfitting, we'll employ early stopping based on validation loss or a composite metric. E.g., if after some epoch, val perceptual loss or val segmentation accuracy starts to worsen, we stop. With few subjects, we might not have a big separate val set, but in cross-val, in each fold we can hold one subject as val while training on others, then test on another. Or simply watch training vs augmented data.

7.4 Curriculum or Staged Training: This can greatly help with stability: - **Stage 1: Warm-up on simpler losses.** For example, first train the model with only L1 and maybe a little perceptual loss, *without* adversarial or topology loss, until it learns a reasonable mapping (e.g., reaches decent PSNR). This avoids the model being overwhelmed by too many objectives at start. In the MDPI stroke paper, they did exactly this: first 60 epochs pixel losses only [48] . We can adopt similar. Specifically, we might *not include segmentation loss for the first few epochs* or set its weight low. This is because if the image is very poor initially, segmentation loss is meaningless (it will just penalize everything). - **Stage 2: Introduce topology loss and/or adversarial.** Once the pixel loss plateau is reached (or after X epochs), we start ramping up the segmentation loss weight. The model at this point has learned to produce something close to 7T in appearance; now segmentation loss will fine-tune it to correct subtle errors in structure. We can do a linear weight ramp or just turn it on fully at epoch N. Similarly, if using a GAN, perhaps start adversarial loss after the model has some baseline quality (this avoids GAN causing divergence early). This approach of gradually adding objectives is known to stabilize multi-objective training [48] . - **Stage 3: Fine-tune on full objective.** After everything is introduced, train until convergence. Possibly do a lower learning rate fine-tuning for final epochs to refine details. - Optionally, **Stage 4: (if diffusion)** we could fine-tune the model as a deterministic generator (like the network weights from diffusion used in a standard UNet) with an L1 loss for a few epochs to directly minimize error. But likely unnecessary if diffusion is done right.

We will design triggers for stage transitions. For example: - "Transition to Stage 2 when validation SSIM > X" or after fixed number of epochs, whichever first. - Or manually: e.g., 50 epochs stage1, then stage2 onward. This is something we'll note in method details. - We saw a reference where stage progression was triggered by hitting certain PSNR/SSIM goals [48] , which is a clever adaptive approach. We might or might not do that automatically, but we can plan approximate epoch counts.

7.5 Optimizer and Training Hyperparams: - We'll use an optimizer like **Adam** (with perhaps β_1=0.9, β_2=0.999 typical). In MDPI example, they used Adam with separate LR for generator and discriminator [49] . - Learning rate: likely around 1e-4 for generator, 1e-4 for diffusion if used. If GAN, discriminator also ~1e-4. For diffusion specifically, some use smaller LR due to many steps, but we'll experiment. - We may use an **exponential moving average (EMA)** of model weights (common in diffusion training) to stabilize and get a slightly better final model [49]. This means maintain a copy of the model that is an EMA of training weights (with decay ~0.999), and use that for sampling or final results – it often yields cleaner outputs. - **Batch size:** Possibly around 4–8 patches per batch depending on patch size and model complexity, to fit in memory. We'll try to maximize batch size to stabilize training (bigger batch smooths gradients a bit), but given 3D patches, memory is a limit. - **Number of iterations:** Because we have effectively an infinite stream of patches, we won't count epochs in the usual sense (since one "epoch" if defined as going through all possible patches is not meaningful). Instead, we'll define a number of training iterations. Roughly, if each subject yields, say, 1000 patch samples per epoch (just by random sampling), then an epoch is 10k samples. We might train for the equivalent of, e.g., 100 epochs = 1,000,000 patch samples. We'll monitor when losses converge or when validation metrics stop improving.

7.6 Failure Patterns and Detection:

During training, we will watch for: - **Overfitting:** If training loss keeps decreasing but validation loss (on held-out subject patches) bottoms out then rises, that's classic overfit. Given heavy augmentation and small network (if we keep parameters in check) plus segmentation reg, we hope to push overfitting limit. But if we see it, we could: - Increase augmentation diversity. - Stop early. - Potentially reduce model capacity (fewer channels or layers). - **Mode collapse (in GAN case):** If we used a GAN and see that the generator starts producing nearly identical outputs for different inputs (a sign of collapse) or the discriminator loss goes to zero (discriminator overpowering generator). To detect: visually inspect a variety of outputs or check output variance. If collapse, remedy by lowering adversarial weight, using regularization like spectral norm, or using diffusion instead of GAN. - **Oscillations:** Especially with GAN or with too high learning rate, losses might oscillate wildly. We would then reduce LR or add gradient clipping (the MDPI paper clipped grad norm to 5.0 [49] which we can also do to avoid rare huge gradients). With diffusion, oscillation is less likely if properly set. - **Loss imbalance:** If one loss term is orders of magnitude larger than others, training might bias towards optimizing it while neglecting others. We will monitor the magnitude of each loss component. For example, adversarial losses are often much smaller numeric values compared to pixel loss; if not scaled, the generator might ignore them. So we ensure to multiply losses by appropriate weights to have them on roughly comparable scale. Similarly, segmentation loss (being a classification loss) might produce values in a different range than pixel loss, so adjust accordingly. - **Output quality issues:** We should inspect intermediate outputs periodically. For diffusion, we might not have an output until we sample after training, but we can generate from time to time using current model to see how images look. For GAN, we can always run the generator on some validation input and see output. If we notice issues like: - Blurriness (means possibly too high L1 weight or insufficient perceptual/adversarial). - Noise or speckle (maybe adversarial weight too high or model capacity issues). - Anatomical mismatches (like a sulcus missing or extra): that's precisely what segmentation loss should catch. If we still see it, maybe segmentation branch not effective enough or weight too low. - Over-sharpening or hallucinated edges (could happen if adversarial or perceptual overshoots): then maybe ease those losses. - Hippocampus asymmetry error (PLOS noted significant asymmetry in hippocamp segmentation with their method [31] [32]). We will specifically check if our synthetic images yield symmetric structures if input was symmetric (healthy brains should be roughly symmetric anatomically). If we find the model, say, made left hippocampus smaller than right unpredictably, that's a problem. Segmentation loss across hemispheres should catch it if the ground truth was symmetric. If not, we could add a small symmetry constraint ourselves (maybe too niche). - **Training time issues:** Diffusion models can be slow to train. With 3D data, if it's taking too long, we might simplify network or reduce time steps. We plan early experiments to calibrate this. - **Convergence plateaus:** If training seems stuck (no improvement for long time), possibly learning rate too low or the task too hard initially. Then we might need to adjust. For example, we might find diffusion is not learning well from scratch on 10 subjects – maybe it needs a pretraining or a simpler approach first (like first train a deterministic UNet with L1 then use it to initialize diffusion model).

We will maintain logs of these metrics and possibly use a tool (TensorBoard or similar) to observe training curves and sample outputs. Because the dataset is small, visual inspection of outputs for each subject is feasible and should be done to ensure no obvious failures on any case.

7.7 Handling Diffusion Specifics (if chosen): Just to clarify, if using diffusion: - We need to choose number of diffusion steps (T). We might use a smaller T like 100 or 200 for efficiency (since 1000-step DDPM might be too slow for 3D). Many modern diffusion use noise schedules that allow fewer steps (or use Denoising Diffusion Implicit Models for fewer steps). We might try a DDIM sampler at inference for speed. But training still goes through T steps each sample – heavy. - Alternatively, consider a **latent diffusion**: encode 3D image with autoencoder to, say, half resolution or smaller, then diffusion in latent space. This would drastically reduce compute. If time allows, we might implement a small 3D VQ-VAE or autoencoder to compress volumes (like compress 0.65mm voxels to 1.3mm or something), do diffusion there, then decode. This is quite advanced but might be necessary for speed. For blueprint, we mention the idea as a footnote if needed, but not commit unless required. - If diffusion is too unwieldy, we'll fall back to the advanced GAN (Candidate B style). That's our contingency.

7.8 Early Failure Detection: We will detect if something is going wrong by: - Monitoring the **segmentation accuracy on a validation subject's synthetic output**. If after some training it's not improving or very low, then the network might be focusing on wrong thing. - Also monitor **image similarity metrics (PSNR/SSIM)** on validation. We expect them to climb initially. If they saturate at a low value (meaning model not learning properly), maybe architecture or training setup needs revision. - Check **discriminator loss trend** (if used): if D loss goes to nearly 0 and G loss stagnates, G is failing (D too strong). If D loss goes high and stays high, D might be too weak or not learning (maybe G dominating or LR issues). Ideally, they should reach some equilibrium range.

In summary, our training strategy is **iterative and vigilant**. We combine multiple loss functions (pixel, perceptual, topology) and possibly multi-task learning to guide the model. We implement a curriculum to introduce complexity gradually. And we track various indicators to adjust course early if needed. This disciplined approach will maximize our chances of the model converging to a solution that satisfies both low-level image fidelity and high-level anatomical correctness.

8. Evaluation Strategy (Reviewer-Driven)

Evaluation is where we convince reviewers that our method is effective. We need to compare against baselines, report standard metrics, and also demonstrate clinical relevance through novel metrics. We will design a comprehensive evaluation scheme as follows:

8.1 Mandatory Baselines: To validate our method's improvements, we will include several baseline comparisons: - **Input 3T vs. Real 7T:** First, as a reference, we'll measure how far a raw 3T image is from the 7T (in terms of metrics like SSIM, etc.). This establishes the "gap" that our model is trying to bridge. It also serves as a naive baseline: using the 3T itself as the output (i.e., do nothing). If our method is any good, it should significantly outperform this identity baseline. - **Linear Interpolation + Intensity Matching:** A simple baseline is to upsample the 3T image to 7T resolution and maybe adjust intensities to match 7T's histogram. Essentially, this is like a classical approach: do high-resolution interpolation and perhaps a bias correction. We can implement: upsample 3T from 0.8mm to 0.65mm using spline interpolation, then perform histogram matching of 3T to 7T (or scale to have same mean/variance as 7T). This will produce a pseudo-7T. We expect this to improve resolution slightly but it won't introduce new details or contrast, so it will likely have lower contrast and missed fine structures compared to real 7T. But this baseline is important to show the advantage of learning-based enhancement. - **Prior Deep Learning Methods:** - *GAN baseline:* We will implement or use a standard conditional GAN (Pix2Pix-style) with a 3D UNet generator and PatchGAN discriminator, trained on our data with just L1 + GAN loss (no topology loss). This represents the type of approach many earlier works used [10]. We expect our method to outperform it, especially on anatomical metrics, but we need to show that. If one of the earlier published methods has code (e.g., Qu et al. 2020 wavelet WATNet), we could attempt to run it on our data. But replicating their custom wavelet network might be heavy. Instead, using a general Pix2Pix3D or even a 3D U-Net with only L1 loss could be two baselines: - U-Net + L1 (no GAN): to show what a pure CNN regression gives (likely smooth result). - U-Net + L1 + adversarial (Vanilla GAN): to show what adding GAN does beyond the above. - *Diffusion baseline:* If our method uses diffusion + special constraints, a baseline would be diffusion without those constraints. For example, conditional diffusion trained purely to minimize pixel loss (no segmentation loss). This would measure the effect of our topology guidance. If diffusion baseline is not easily trainable, at least we have the GAN baseline. But since diffusion is our chosen main, it's valuable to show how adding segmentation loss improves metrics vs plain diffusion. We can do an experiment: train a diffusion model with just conditioning and compare. - *From literature:* If any known method has reported results on this same dataset (given the dataset is new 2023, possibly not many have published on it yet, except maybe that PLOS 2025 which did 7T synthesis on a different dataset). We could consider comparing with results from Qu et al. (2020) if they reported numbers like PSNR/SSIM on their data. Or the MICCAI 2021 (Do et al. 2021) if available. However, lacking direct access or code, we will rely on implementing basic versions ourselves as above. - *Ablation (counts as baseline):* We will treat our own model with certain components ablated as baselines too. For example: - Ours minus topology loss (i.e., remove segmentation loss). - Ours minus diffusion (if we ended up with diffusion, compare to if we did a similar architecture with standard training). - Ours minus T2 input (if we included T2, show result with and without it). These are not external baselines but internal comparisons to justify each component. Reviewers expect ablations to prove that each novelty (topology constraint, etc.) has a measurable effect.

- **Human Expert Baseline:** If possible, we might do a qualitative comparison: e.g., have an neuroradiologist or experienced viewer compare images or do a Turing test – can they distinguish real 7T from synthetic? If we can claim that synthetic images are almost indistinguishable in blind tests, that’s powerful (the SWI diffusion paper did statistical tests showing no significant difference in quality scores [25]). If we have access to an expert or can do a small user study (maybe with 2-3 imaging scientists rating image pairs), we will include those results. This is optional but would strengthen the evaluation of perceptual realism.

8.2 Quantitative Image Quality Metrics (Technical Metrics): We will compute standard metrics on the test set comparing synthetic 7T to real 7T: - **PSNR (Peak Signal-to-Noise Ratio):** in dB. This is common, though it correlates poorly with perceptual quality sometimes. Still, a necessary metric to report because many prior studies do [50]. Higher PSNR means closer to ground truth. We expect our method to have higher PSNR than baselines (especially input or naive). - **SSIM (Structural Similarity Index):** This measures perceived image similarity accounting for luminance, contrast, structure. It’s more informative than PSNR for structure preservation. We’ll report average SSIM over the brain region. - **NMSE (Normalized Mean Squared Error):** Some papers report NMSE or MSE. It’s basically 1 - (PSNR in a way). We might include NMSE since PLOS did [28], to compare with possibly their numbers or just as another view. - **FID (Fréchet Inception Distance):** Possibly we could compute FID between synthetic and real 7T distributions. However, FID requires an "Inception" network trained on similar images. We might try using a 3D CNN classifier's features (or 2D slices and use a 2D Inception). FID can capture if synthetic images as a set differ in distribution from real. If we have enough samples (we have small number though). With 10 images, FID may be unstable. But if we consider many patches, maybe. We might skip FID due to small N. Another distribution metric could be **kernel MMD** between synthetic and real image patches, which TopoGAN used for topology but could also be used for intensity distribution. - **Perceptual Score:** We could use learned perceptual image patch similarity (LPIPS) metric (there's a standard implementation using VGG features). This would quantify similarity in feature space. If our method yields lower LPIPS (closer to 0 means very similar), that's good. We might include it to emphasize perceptual closeness beyond pixel similarity.

We will likely tabulate these metrics for: - Each method (ours and baselines) on the test set. - Possibly broken down by region or something? Usually, one reports global brain metrics. But we might also compute error metrics regionally: e.g., PSNR specifically in gray matter vs white matter (because maybe model does better in one than other). We can do this by masking the images with tissue segmentations and computing MSE in those areas. This could be an interesting analysis: perhaps our model significantly improves gray-white contrast (so better SSIM in gray matter areas). - We'll apply statistical tests on these metrics if appropriate (with 10 subjects, a paired t-test or Wilcoxon between ours and baseline metrics can show significance). For example, compare SSIM of our method vs SSIM of baseline, get p-value. That is commonly expected in journals: to say improvement was statistically significant ($p < 0.05$).

8.3 Metrics that Matter Clinically: Here we go beyond pixel similarity to show that our synthetic images preserve or enhance clinically relevant information: - **Segmentation Accuracy:** Using the provided ground truth segmentations (from FreeSurfer on real 7T) as reference, we will run the same segmentation tool (or our own segmentation head) on the synthetic 7T and measure accuracy. For example, compute **Dice coefficients** for major brain structures (white matter, cortex, ventricles, hippocampus, amygdala, etc.) between segmentation of synthetic and segmentation of real 7T [28]. If our synthetic image is truly anatomically faithful, these Dice scores should be very high (close to 1). Baselines likely yield lower Dice (especially in tricky structures). The PLOS paper did something similar by evaluating differences in volumes from VolBrain segmentation [28] [32]. We will essentially replicate that approach: treat the real 7T segmentation as ground truth, and assess how well one would segment the synthetic vs how well one segments the real 3T (as a baseline). - E.g., measure hippocampal volume error: |Vol_syn - Vol_real| relative to |Vol_3T - Vol_real|. If our method is good, the synthetic's hippocampal volume should match the real 7T's within a small margin, significantly better than just using 3T (which might systematically underestimate CA1 region as per Kerchner's note [14]). - We'll do this for multiple structures. Key ones for AD: hippocampus, entorhinal cortex, ventricles (since enlargement indicates atrophy), possibly whole brain volume. - **Topology-specific Metrics:** We can adopt metrics inspired by the topology-preserving literature: - **Euler Characteristic or Betti numbers:** For example, compute the Euler characteristic of the binarized white matter mask in real vs synthetic. If synthetic wrongly breaks white matter into two pieces, Euler char will differ. We expect identical Euler char for brain compartments. We could do this for multiple threshold levels to mimic persistent homology but simpler. - **Connected Component Similarity:** For instance, ensure that the number of connected components in the segmentation of synthetic matches that in real for key structures (cortex should be one connected sheet per hemisphere, etc.). We could say we checked and found no extra components. - **Boundary Distance:** Compute the Hausdorff distance or average surface distance between boundaries of structures in synthetic vs real. For example, take the hippocampus segmentation from real vs synthetic, compute distance between surfaces; a small mean distance indicates shape is well preserved. This quantifies topology/shape fidelity. - **Persistent Homology Metric (Topological Similarity, TS):** The vascular paper defined a TS using persistent homology overlap [29]. We could attempt a simpler analog: compute persistence diagrams for segmentation volumes and measure their similarity. This may be overkill given segmentation differences/dice cover most needs, but we can mention if any topological invariant deviance was found (likely not, if dice is high). - **Contrast-to-Noise and SNR:** MRIQC metrics used in the Sci Data (CNR, SNR) [40] could be computed on synthetic images to see if they approach real 7T values. For example, total SNR synthetic vs real vs 3T. Or GM–WM contrast ratio. This indicates whether our images truly mimic the improved contrast of 7T. If synthetic has higher CNR between GM and WM than 3T, that's a win (closer to 7T's known high contrast [15]). We can measure CNR by taking mean intensity in GM vs WM segment and dividing by noise std (estimated from air background). We could present that in a small table or text: "Synthetic 7T achieved GM/WM contrast of X, compared to real 7T's Y and input 3T's Z [40]." - **Alzheimer's-related evaluation:** This is critical for our claim about early AD analysis, but since our data are all healthy, we need to be creative: - We can simulate an AD scenario by using known biomarkers. For instance, one hallmark is hippocampal atrophy. On 7T, one might delineate hippocampal subfields better [14]. We can do a test: *Add artificial atrophy* to the 3T images and see if our model still preserves/detects it. For example, take one subject's 3T, intentionally shrink the hippocampus region slightly (by masking and reducing intensity or something), feed through model, see if output shows a volume reduction in hippocampus compared to normal synthetic. This is artificial, but demonstrates sensitivity. It might be too contrived though. - Alternatively, leverage a separate dataset: if we have any 3T scans from AD patients and maybe those same patients had 7T (rare) or at least 7T from other AD vs control? Possibly unrealistic. We could however use our model on actual AD 3T scans (from ADNI, etc) and feed synthetic images into an AD classifier. If we had a pre-trained classifier on MRI (like one from literature that takes MRI and outputs probability of AD), we could compare classification on original 3T vs synthetic-7T (which presumably has higher contrast). The hypothesis: the classifier might perform better or give higher confidence for AD on the synthetic because subtle features stand out more. This is a bit speculative and requires careful setup (domain shift might confuse classifier). If we have time, we might do a small experiment: train a simple CNN to classify AD vs control on some dataset of 3T MRIs (like ADNI 3T), then test it on ADNI 3T images processed with our model (synthetic "7T") and see if any difference in prediction or feature activation. This is a heavy evaluation so maybe just a discussion point if not fully done. - More directly, we can measure **effect size of group differences**: e.g., known that hippocampal volume difference between AD patients and controls is X% at 3T. If we *had* any AD vs control images to run through (maybe from ADNI, albeit no ground-truth 7T for them), we could see if synthetic images yield a larger separation in volumes, suggesting earlier detection. Without AD data, we cannot do this with our 10 subjects (all healthy). We might skip actual AD testing due to lack of data, but stress that we preserve the metrics that matter (like hippocampal volume) which correlate with AD.

Given limitations, our main "clinical" demonstration will likely revolve around showing **structural measurements** (volumes, thickness) from synthetic images are nearly identical to those from real 7T. Real 7T presumably is a gold standard for measuring those. If synthetic replicates them, it implies one could get 7T-level measurement accuracy from 3T images. For early AD, volume of CA1 region as measured on 7T was found to correlate with mild AD [14]. If our synthetic can yield a similar measurement, that's a success. We will highlight that qualitatively: maybe show a plot of hippocampal volumes: 3T segmentation underestimates volume, 7T truth is higher, synthetic matches the 7T (assuming 3T had partial volume effects, etc.). If we can generate such evidence, it directly addresses AD angle.

8.4 Statistical Testing: We will apply statistical analysis to our results to establish significance: - For each metric (PSNR, SSIM, dice, volume error, etc.), we will do paired tests comparing our method vs baseline on each test subject. For example, for each subject, compute SSIM(ours) - SSIM(baseline), then do a paired t-test across subjects to see if mean difference > 0 significantly. Given N=10, if differences are large, it might reach significance ($p<0.05$). If not, we'll report "no significant difference" honestly for that metric. - For segmentation volumes, we might do a similar test on absolute errors or dice scores. - If distributions are not normal (with 10, hard to tell), we could use non-parametric Wilcoxon signed-rank test. But t-test is probably fine and expected. - If we do cross-validation and get multiple values, we might average them for final, but significance can still be assessed by pooling or by an appropriate test.

We should also test if certain aspects are significantly improved: - For instance, is the improvement in hippocampal volume error statistically significant, etc. - In PLOS, they highlighted hippocampal asymmetry as a failure. If our method reduces that error, we'll mention it.

8.5 Ablation Studies (minimum required): We plan to perform ablations to isolate contributions: - **Without Topology Loss:** Train the same model but without the segmentation/topology loss. Compare metrics, especially segmentation accuracy and connectivity. We expect: without topology loss, the model might have slightly higher PSNR (since it can freely adjust intensities) but likely worse segmentation fidelity. For example, maybe hippocampus dice drops or a subtle structure is missed. We will report those differences. If results show negligible difference, reviewers might question our complexity; but we suspect differences will be there (prior work indicates these constraints matter for fine details [31]). - **Without Multi-Modal (T2):** If we included T2 in main model, we should ablate it: train the same model using only T1 input. Then see if adding T2 gave improvement. If yes, we quantify that (say SSIM improved by X, or better detail in some region). - **Without Diffusion / Using GAN:** This one is trickier since it's a wholly different approach. But we can ablate the generative method: e.g., "If we remove the diffusion probabilistic aspect and train the network deterministically with the same losses, results are Y." Or vice versa: "If we remove adversarial loss and only use diffusion, results are Z." Essentially, we want to justify using diffusion (if we did). If our final method is diffusion, we can compare to a GAN baseline which is a form of ablation at method-level. Or if our final ended up being a hybrid, we can try turning off adversarial part, etc. - **Different architecture choices:** Possibly not needed to show to reviewers unless it's a major question like "why transformer?". If we did use a transformer and it helped, we could ablate it by replacing with a plain UNet to show maybe a small drop in performance. If our results are clearly strong, this might not be necessary to explicitly show, but we can mention it in discussion.

We'll include at least the first two ablative comparisons in the results section, likely as part of a table or separate sub-table, showing the effect on metrics. Additionally, we might include qualitative figure comparisons for ablations: e.g., show a zoom on a hippocampal region for: - ground truth 7T, - our full method, - our method without topology loss. Perhaps the one without topology might have a slightly blurred hippocampal outline or different shape, whereas ours preserves it. Visual proof plus metric differences would be convincing.

8.6 Publishable Novelty vs Incremental: We explicitly frame what in our results constitutes *novelty*: - If our method shows significantly better preservation of fine anatomical detail (like we can show synthetic 7T allows segmentation of a structure that was impossible with 3T), that's a **publishable outcome**. For example, perhaps 7T can reveal the hippocampal subfield CA1 vs CA2 boundary. If our synthetic images allow a subfield segmentation algorithm to segment CA1 vs CA2 similarly to real 7T, that is novel and clinically significant (since 3T could not do that). - We will emphasize any result that shows *improved clinical metric*. E.g., "Using synthetic 7T, the error in hippocampal volume measurement was reduced from 8% (with 3T) to 2%, approaching the 7T reference." This indicates a practical improvement for researchers measuring atrophy – exactly what we claim for AD. - Also, if we demonstrate improved *detection of subtle lesions* (maybe one of the references was that 7T reveals microbleeds and microinfarcts [51] , which 3T might miss. If our synthetic can simulate those? Hard because our healthy data probably has none. But if a microbleed was in 7T, can model put it in synthetic? If not explicit, we might discuss potential). - We'll highlight the **topological consistency**: e.g., "Unlike baseline GAN which occasionally produced disconnected cortical fragments or artifactual holes, our method maintained identical topological structure, verified by identical Euler characteristics and high Dice scores across all structures." This directly addresses the novelty of topology preservation.

What would be considered **incremental** by reviewers? If our results only showed marginal improvement in PSNR or a small visual improvement with no clear impact on segmentation or analysis, they might say it's incremental. We guard against that by focusing on the big differences: - For instance, if baseline U-Net had SSIM 0.90 and our method 0.92, that's nice but not groundbreaking. But if we show baseline had hippocampal Dice of 0.80 and ours 0.90, that's more meaningful because it ties to structural fidelity. - So we expect our novelty to be "the first method to explicitly enforce and demonstrate preserved anatomical fidelity in 3T→7T synthesis, leading to significantly improved downstream measures (like segmentation accuracy approaching real 7T, and better preservation of disease-relevant structures)". That is our key selling point to make sure it doesn't read incremental.

8.7 Downstream AD Evaluation Plan: We should articulate how we evaluate the images for AD-related tasks: - If possible, do at least a case study: take one of the references that 7T visualized something in an AD patient that 3T didn't [14]. Perhaps we can simulate that scenario with our synthetic image. For example, reference Kerchner 2010: they saw specific atrophy in CA1 region on 7T in mild AD. We could take a mild AD 3T scan (from ADNI) that presumably has subtle atrophy, run our model to get synthetic 7T, and then see if a neuroradiologist can more easily identify the CA1 thinning on the synthetic vs on original 3T. This is anecdotal but could be included as a figure if possible. - If not, at least measure if synthetic images show increased contrast in the hippocampal layer that could correspond to that finding. For example, measure the difference in intensity between hippocampal subfield vs surrounding on synthetic vs 3T. - We might also incorporate a **neurodegeneration score**: maybe compute whole brain vs ventricle volume ratio (an atrophy metric) on synthetic vs 3T. If a patient had slight brain atrophy, 7T's clearer CSF boundary might measure it more accurately. If synthetic replicates that, it shows better detection of atrophy.

Given the constraints, we'll probably not have actual AD ground truth to quantitatively show improvement. Instead, we show that *if one used our synthetic images for the same analyses one would do on 7T for AD, you get virtually the same results as real 7T.* This implies one can achieve the benefits of 7T in AD research without actual 7T. That itself is a strong statement if evidenced by segmentation and morphometry results.

We will clearly state what qualifies as success: e.g., "We consider our approach successful if it significantly outperforms baseline image translation methods on standard image quality metrics **and** if it yields no significant differences in key anatomical measurements compared to real 7T images. A publishable novelty is demonstrated by bridging the gap between 3T and 7T in terms of these measurements – for instance, achieving segmentation accuracy on synthetic images that is on par with using real 7T (with any differences statistically insignificant [25]). Simply showing a slight bump in PSNR, on the other hand, would be considered incremental."

Finally, we'll prepare visual results: - **Figures:** likely an array of slices for qualitative comparison (3T vs our synthetic vs real 7T vs maybe baseline's synthetic). Reviewers always want to see actual images to judge if improvements are noticeable. We'll ensure to include zoom-ins of critical regions (cortex or hippocampus) to highlight differences (maybe mark where baseline fails e.g. baseline blurred a small gyrus that ours preserves). - Possibly a figure showing segmentations overlay: e.g., outline of hippocampus from real 7T segmentation overlaid on synthetic image to show it aligns well, whereas on baseline's output it might not align due to shape distortion.

All these evaluation components together will make a strong case that our method not only produces pretty images, but images that are *quantitatively as useful as real 7T* for neuroanatomical and potentially diagnostic purposes. That's key for Q1-level impact.

9. Reproducibility & Research Hygiene

Top-tier journals and conferences increasingly emphasize reproducibility and rigorous experimental practices. We will adhere to these expectations by enforcing discipline in experiment tracking, data management, and reporting:

9.1 Experiment Tracking and Version Control: - We will use a tool like **MLflow or Weights & Biases** to log every experiment's parameters, code version, and results. This ensures that for each run (with specific hyperparameters or architecture changes), we have a record of what produced the results. It will help us later when writing the paper to cite exact numbers and to retrieve models for qualitative results. - Each model configuration (e.g., "Baseline U-Net L1" vs "Our method full") will be assigned a clear name and possibly saved as a trained model checkpoint with a version tag. We'll use Git for code management, committing changes with descriptive messages (especially for changes affecting results). - We will fix random seeds for all critical processes (data shuffling, network initialization) whenever we report final results, to ensure replicability. For instance, when computing metrics for paper, we might run with 5 different seeds and average, but we'll always note the seed or range so results can be reproduced. - The dataset being BIDS, we will specify exactly which files were used (like "aligned NIfTI images from Figshare, version DOI:..."). This ensures someone else can download the dataset and follow our steps.

9.2 Data Splits and Cross-Validation: - We'll clearly define how data is split. For example: "We performed 10-fold leave-one-subject-out cross-validation. In each fold, 8 subjects were for training, 1 for validation (to monitor training), 1 for testing. This way, every subject appears in test exactly once." We'll provide a table or list of which subjects were in test in each fold (maybe in supplement if needed). This transparency lets others know we didn't cherry-pick a favorable split. - We will maintain the same splits for comparing all methods. That is, baselines and our method will be evaluated on the identical test sets for fairness. We won't, say, show our method on easy cases and baseline on hard. - If we decide not to cross-val and instead hold out e.g. 2 subjects as final test, we'll identify them (like sub-09 and sub-10 were test) and make sure validation was only used for hyperparam tuning, not for model training. But likely cross-val given small N.

9.3 Parameter and Implementation Details: - We will list key hyperparameters in the paper (and definitely in any supplemental material). E.g., patch size, batch size, learning rates, number of training iterations, diffusion timesteps, loss weights, etc. We'll also mention the hardware used (like "Trained on an NVIDIA A100 GPU for 3 days" etc.) so others have an idea of computational requirements. - If we use any external software (FSL, FreeSurfer, etc.), we'll specify versions and settings (like "FreeSurfer v7.1.0 with default recon-all used to obtain surfaces [34]"). - All random aspects (like data augmentation) will be described sufficiently (range of rotation, etc.) so someone could replicate that environment.

9.4 Source Code Release: - We plan to release our code (likely via GitHub) upon publication (or even submission if allowed). This includes training code, inference code, and perhaps our pre-trained model weights. Releasing code is increasingly expected for deep learning studies, especially if we claim state-of-art performance. It allows reviewers and readers to verify claims. We'll ensure the code is cleaned up and well-documented by the end. - If any part of code is tricky (like computing a topological loss), we might provide a snippet or clear explanation in the appendix as well.

9.5 Reproducible Training: - We will fix random seeds for final model training runs so that results can be exactly reproduced. For diffusion, there's some inherent randomness in sampling; we'll mention that but also how to fix it (seed the sampler). - We will ensure that our training pipeline yields the same result if run twice (within minor randomness differences) by controlling nondeterministic operations as much as possible (PyTorch can be set to deterministic convolution, etc., albeit with performance cost). - We might also use docker or an environment YAML to specify all package versions. Perhaps provide a Dockerfile or Singularity container for others to easily run.

9.6 Reporting Standards: - In results, we will not cherry-pick the best output examples only; we'll show average performance across subjects and include variability (std or error bars). For qualitative, we might choose representative slices but will state that they are representative. - We will follow guidelines like the **ISMRM Reproducible Research Checklist** or MICCAI's checklist: e.g., mention how many times experiments were repeated (if relevant) to ensure stability, whether results are median or mean of multiple runs, etc. - For deep learning, **we will provide confidence intervals** for key metrics. With only 10 test samples, a non-parametric CI (like bootstrapping across subjects) can be given. That signals statistical rigor.

9.7 Handling Negative or Mixed Results: - If an aspect of our method didn't help (say we tried a transformer and it gave no improvement), we won't hide that – we can mention it either in ablation or discussion. Honesty about what did not work as expected actually improves trust in the research. We can explain perhaps why it didn't help (maybe due to small data, etc.). This helps others not go down the same wrong path or understand limitations. - Also if our method has failure cases (maybe one subject had unusual artifact where our method failed), we will mention that and perhaps show it in supplement, to be transparent.

9.8 Data and Model Availability: - The dataset is public (Figshare), and we'll cite its DOI [34]. We'll clarify any pre-processing we did and possibly share derived data (like the exact aligned & normalized volumes we used) if license allows. Since it's open data, sharing processed versions should be fine as long as we cite original. - We intend to share the **trained model weights** for our final model. This would allow others to apply our model on their 3T MRI directly, which is valuable for adoption. We'll host it on a repository or in the paper's supplementary. - We'll also provide instructions to run inference (like a script where you give a 3T NIfTI and it outputs synthetic 7T NIfTI). - If space permits, we might include a small table of *hyperparameters used* in an appendix, to avoid any ambiguity.

9.9 Top Reviewers' Reproducibility Expectations: Reviewers in 2025 often look for: - Open source code (preferably with a permissive license). - Clear description of train/val/test splits (which we will provide). - Possibly registration of the study (less common in imaging, more in clinical trials; not needed here). - Evidence that results are not due to one lucky run (some require multiple random seeds; given small data, we might at least mention if similar performance was observed across different initializations or if we ensemble models). - They may check if any learning-based method could be heavily dependent on initialization. We can show low std across seeds, or mention we trained multiple times and got similar metrics (if true).

We will include statements like: “We ran each model training 3 times with different random initializations; the variation in SSIM was <0.005, indicating stable training.” This assures them reproducibility in the sense of not being a fluke result.

9.10 Ethical and Bias Considerations: Even though not explicitly asked, reproducibility includes acknowledging limitations. For instance, our data is 10 healthy young adults (3 female, 7 male) [52]. We should note this demographic and caution that results might differ on older or diseased populations (if, say, AD brains have different characteristics, the model might need fine-tuning). We’ll stress that further validation is needed on larger, varied cohorts (if we had access to any AD scans with 7T, that would be ideal, but if not, we say as future work). We’ll also mention that our method is not intended for direct clinical diagnostic use yet, but as a research tool to enhance images for analysis, requiring further validation in clinical settings.

9.11 Organized Results and Code: We plan to accompany the paper with a results repository: including the output images for test cases (so others can visually check them), and maybe the metrics per case in a CSV. This level of detail helps if someone wants to do a meta-analysis or compare their method to ours later.

In summary, we will make it as easy as possible for someone else to reproduce our entire pipeline: from downloading the same data, running our preprocessing (we’ll share scripts), loading our model, and getting the same outputs. By doing so, we also insulate ourselves from reviewer skepticism – any claim we make can be verified with the provided materials. This level of transparency and rigor is increasingly demanded by Q1 journals and will strengthen our paper significantly.

10. Paper Strategy (Q1-Focused)

To target a high-impact journal or top conference, we need not only strong content but also a clear and compelling presentation. Below is our strategy for the paper’s structure, emphasis, target venues, and addressing potential reviewer objections:

10.1 Target Venues: - **Journals:** Given the data-driven, neuroimaging focus, **NeuroImage** is a prime target (it’s Q1, widely read in neuroimaging, and has published image enhancement and synthesis works [53]). Also, **Medical Image Analysis (MedIA)** or **IEEE Transactions on Medical Imaging (TMI)** are suitable for methodological rigor combined with medical relevance. MedIA might appreciate the topology-preserving angle and thorough evaluations. If focusing more on the Alzheimer’s angle, **NeuroImage: Clinical** or **Alzheimer’s & Dementia (the journal)** could be options, but they typically prefer more direct clinical results. Our work is methodological, so NeuroImage or MedIA are top choices. Another is **IEEE Transactions on Biomedical Engineering (TBME)** if we emphasize the technical innovation (they published earlier GAN works [10]). - **Conferences:** As interim or alternative, **MICCAI** (Medical Image Computing and Computer-Assisted Intervention) is top-tier for such work, especially if we highlight the technical novelty. MICCAI reviewers would like the topology-preserving contribution and multi-task learning aspect (provided we show strong results). **IPMI 2025** (Information Processing in Medical Imaging) is another specialized high-quality venue that values rigorous methodology – it’s smaller but prestigious. Given we want a Q1 journal paper, we might first submit to a journal directly. If time aligns, maybe a MICCAI submission as well and then expand to journal.

10.2 Paper Structure: We’ll follow the IMRaD structure with enhancements: - **Introduction:** 1) Introduce the problem: 7T vs 3T gap, importance for AD (cite those studies showing 7T benefits [1]). 2) State the unmet need: not everyone has 7T, hence the desire for 3T-to-7T synthesis. 3) Summarize limitations of existing approaches (GANs can improve visuals but may not guarantee anatomical fidelity [31]; mention no one has addressed topology explicitly and the risk of hallucination). 4) Summarize our solution approach (diffusion + topology constraints, multi-modal, etc.) and why it addresses those limitations. 5) Contributions bullet list: e.g., "(i) We introduce a topology-preserving loss for MRI synthesis that maintains anatomical correctness; (ii) We demonstrate on a public paired dataset that our method achieves significantly better structural fidelity than prior methods; (iii) We evaluate the impact on downstream measures (segmentation accuracy, etc.), bridging the gap to real 7T's clinical utility." This sets the stage and claims novelty clearly. - **Related Work:** We'll have a section to discuss previous 3T->7T methods (Zhang 2019, Qu 2020, Do 2021, Park 2024 (GAN-MAT) for cross-modality, etc.) [9] [54] [26]. Also mention works on topology in medical images (TopoGAN [27], segmentation topological losses [55]). Also diffusion in medical imaging (point to SWI paper as evidence of diffusion rising [13]). We will position that none addressed our exact combination of problems (no one targeted early AD specifically or guaranteed segmentation fidelity in the synthetic images). - We also ensure to cite the dataset paper [56] to show we used it appropriately. - If transformer used, mention emerging use of ViTs in med imaging. - **Methods:** We will likely break it into subsections: 1. **Problem Formulation** – define input/output formally (maybe not needed if straightforward). 2. **Preprocessing** – briefly describe alignment, etc., possibly in an appendix if space is limited. But in a journal, we can detail it (this covers our Section 5 in condensed form). 3. **Model Architecture** – describe the generator (diffusion or GAN) and any special blocks (transformer). Possibly include a diagram (the one described in Section 4) as a figure. Label the important parts like multi-task outputs. 4. **Loss Functions** – equations for our composite loss (e.g., L_total = λ1 L1 + λ2 L_percep + λ3 L_topo + ...). If diffusion, describe how we incorporate conditioning and segmentation loss. 5. **Training Procedure** – describe patch sampling, augmentation, curriculum (these details show we did it carefully). Might be a separate subsubsection. 6. **Baselines** – describe any implemented baselines for fair comparison (so in Experiments section, readers know what "U-Net GAN" refers to). All method descriptions will be precise enough that a knowledgeable reader could implement them (with code release making it easier). - **Experiments/Results:** - **Dataset**: describe the paired dataset (10 subjects, what sequences, etc.) [5] [39]. Possibly provide a small table of subject info (age range, etc.). - **Experimental Setup**: explain cross-val, augmentation details, hardware used, training time, hyperparams chosen. This part ensures reproducibility. - **Evaluation Metrics**: explicitly define PSNR, SSIM, etc., and also define our segmentation metrics, etc. We'll mention we use FreeSurfer segmentation as ground truth and measure dice, etc. - **Results – Image Quality**: Present quantitative table comparing methods on PSNR/SSIM/NMSE/LPIPS. Use bold to highlight best scores. Do statistical significance markings if applicable. - **Results – Anatomical Fidelity**: Present another table or figure for segmentation-based metrics. E.g., a table of dice for key structures (maybe a row for whole brain, GM, WM, hippocampus). Or a bar chart showing volume errors for baseline vs our method. - **Qualitative Results**: Provide one or two figures. One could be multi-slice view of an entire brain (to show general appearance is good); another figure zooming into a detailed region to highlight differences (like we might show: 3T vs synthetic vs real, plus an error map perhaps). Could also overlay edges or segmentation boundaries for visual compare. - **Ablation Results**: Possibly a small subsection "Ablation Study" where we say "Removing the topology loss resulted in X drop in hippocampal Dice (Table Y) and qualitatively led to slight discontinuities (Fig Y)". We might compress ablation numbers in text or a small table. - **Downstream Task Evaluation**: If we did any classifier or group difference test, present that. Possibly: "We fed the images to an AD classifier: classifier accuracy on synthetic vs real vs original compared." Or "For each subject, we measured GM volume; the correlation between synthetic and real 7T volumes was R=0.99, whereas between 3T and 7T was R=0.95, indicating improved measurement accuracy." - We'll make sure each claim in introduction is backed by a result here. For instance, if intro claims "we preserve topology", the results section must explicitly show evidence (like "no topological errors observed, segmentation dice ~0.99 for all structures", etc.). - **Discussion:** - Start by reiterating key findings: e.g., "We achieved near-real-7T image quality and demonstrated for the first time that a synthetic image can be used interchangeably with a real 7T for segmentation tasks." - Compare with other works: perhaps "Qu et al. reported a SSIM of X in their 2020 paper on a similar task; our SSIM is higher by Y% while also preserving segmentation accuracy" – if applicable. Or "Unlike prior GAN approaches [31] which reported issues like hippocampal asymmetry, our method maintained symmetry as evidenced by ...". - Highlight the benefit for AD research: "This is important because early AD changes in hippocampal subfields can now be studied using widely available 3T scans enhanced by our method, potentially unlocking new analyses without needing ultra-high-field scanners." We could even hypothesize that our approach could be extended to other field differences (like 1.5T to 3T) to democratize image quality improvements. - Acknowledge limitations: "We only evaluated on healthy subjects due to data availability. While we expect the model to generalize to pathological brains (since topology loss anchors it to anatomy, and intensities could be adjusted via fine-tuning), future work should validate on patients (e.g., ADNI data) and possibly incorporate pathological data during training to avoid any bias." Also mention small sample is a limitation but cross-val and augmentation mitigated it. - If diffusion: mention that sampling is slower than a GAN, but since it's offline enhancement (not real-time), it's acceptable; however, one might explore faster samplers or combining with GAN for speed. - If any failure cases, discuss: maybe our method might have trouble if input quality is very low (motion-blurred 3T) – then synthetic might amplify artifacts. We should caution about garbage-in garbage-out. - Also discuss the concept of hallucination: though we preserve topology, the fine details added by the model are not actual measured data, so any analysis on them should be mindful that those details are model-based. However, since we guided by known structures, we hope it's reliable. But still, it’s good to clarify that clinicians shouldn’t blindly trust a synthetic image to diagnose a new subtle pathology the model wasn't trained for (e.g., a small tumor might not appear if not in training data). - Suggest future improvements: maybe integrating quantitative MRI constraints, or training on larger multi-center data, or extending to other contrasts (like FLAIR). - Possibly mention that the method could be applied to other enhancements (like 7T ex vivo or 7T to even higher? Or 3T to higher resolution at same field). - **Conclusion:** Summarize in a few sentences: "We presented a novel topology-preserving 3T-to-7T MRI enhancement model that produces anatomically faithful synthetic 7T images. It outperforms conventional GAN-based synthesis in both image quality and retention of clinically relevant features. This approach can expand access to ultra-high-field-like neuroimaging and potentially improve early detection of neurodegenerative changes on standard MRI. We will release our code and model to facilitate further research in this direction." Short and impactful.

Throughout each section, we will emphasize novelty: - In Intro and Discussion, explicitly say how we are first or better in certain aspects. - The writing tone: as a research director, it should sound confident, factual, and not overly conjectural. But also objective in acknowledging limitations, which reviewers appreciate.

10.3 Typical Reviewer Objections and Our Preemptive Measures: - *"This is just an incremental combination of known methods."* – We will counter this by clearly delineating that while we build on diffusion/GAN/etc., the specific integration of a topology loss in MRI synthesis is new. We provide evidence that this integration solves a problem others had (like preserving hippocampal integrity that prior art didn’t address). Also, diffusion in this exact context is brand-new (the SWI was recent and different contrast [8]). - *"Small sample size, results not generalizable."* – We preempt by cross-validating, augmenting, and showing the model’s robust performance on each fold. We also might argue that even with 10 subjects, the consistency of results and the strong guidance (segmentation) help prevent overfitting. We’ll mention that adding more data would likely further improve it, and we plan to incorporate more in future, but this dataset was a demonstration. We might suggest that our method could be retrained on larger datasets when available (making it a methodology paper that can scale). - *"Why not evaluate on AD patients to support the early detection claim?"* – We anticipate this. Our defense: obtaining 7T scans of AD patients is extremely difficult (only a few studies have such data). We focused on method development and validated that we preserve the features known to be relevant for AD (like hippocampal volume, microstructural contrast). We could mention that as future work we will test the model's effect on clinical classification tasks once such data can be obtained. Perhaps we can soften any strong claim in the paper about AD: we will say "downstream analysis, e.g., for AD, could benefit, as we illustrated by surrogate metrics" rather than outright "we detected AD better." Keep it realistic. - *"The improvements in metrics X and Y are modest."* – If that happens, we will emphasize the qualitative and structural findings more. However, if improvements are modest but consistent and we have the novelty of topology, we should stress that it's not just about improving PSNR by big margin, but about guaranteeing no errors. For example, "Our method yields slightly higher SSIM than baseline, but crucially, it eliminates the structural errors that baseline had (which SSIM might not fully capture). This is evidenced by perfect segmentation fidelity etc. In critical applications, avoiding any false anatomical changes is more important than a small boost in SSIM." Essentially shifting focus to quality vs quantity of improvement. - *"Could this cause hallucination of pathology that isn't there or hide pathology that is?"* – We will address that in Discussion: our topology loss is supposed to prevent creation of structures that don't exist in 3T. However, subtle pathologies that are below 3T resolution might not appear in synthetic because the model hasn't seen them (if none in training). Conversely, it shouldn't hallucinate a lesion if there's no hint of it in 3T. We'll note that caution is needed if using synthetic images for diagnosis – ideally it should complement, not replace raw data in uncertain cases. This candor shows we are aware of limitations.

10.4 Emphasis in Each Section: - Intro: emphasize big picture and novelty strongly, hook with the idea of bridging 20,000 vs 100 scanners gap [4] (some numbers from Sci Data to catch attention). - Methods: be detailed enough for reproducibility (especially since this is a journal likely). - Results: emphasize not just numbers, but what they mean (e.g., "this Dice score means synthetic vs real segmentations differ by only 2%, which is within inter-rater variability, indicating essentially no information loss in segmentation"). - Discussion: emphasize how this method could be applied or extended, showing it's a line of research that can grow (journals like when authors show how it contributes to field direction).

10.5 Pre-empt Rejection: We will ensure to: - Include a strong motivation and novelty statement early (so reviewers don't think "so what"). - Provide evidence for all main claims (if we claim topology preserving, have a figure or metric showing it). - Be very honest about limitations to gain trust, but also argue that limitations do not invalidate the main claims but rather provide opportunities for future work. - Provide access to code/data to appease any reproducibility sticklers.

Given our thorough plan, we should be able to handle most critiques. If a reviewer is skeptical about clinical significance, we will have enough analysis (like segmentation fidelity etc.) to demonstrate potential. If skeptical about novelty, we lean on the unique combination and what it achieves that others didn't.

By carefully constructing the paper as above, focusing on a clear story (3T->7T possible with no loss of anatomical info, enabling new research) and backing everything with data, we set ourselves up for a positive reception in a Q1 venue.

11. Timeline & Execution Plan

To execute this project efficiently and thoroughly, we will break it into phases with clear goals and decision points. Below is a phased timeline, with approximate durations and criteria for moving forward at each phase. We assume we have no hard deadline, but we aim for a high-quality outcome (which might be roughly a 12-month project given research cycles).

Phase 0: Literature Deep Dive & Preliminary Validation (Month 1) - **Deliverables:** Comprehensive literature review (2021-2025) confirming the novelty of our approach; a list of potential methods and evaluation criteria. Also, a re-analysis of the dataset paper and maybe contacting authors if needed for clarifications. - **Tasks:** Read key papers (the ones we cited and beyond) to ensure we understand pitfalls. Possibly reproduce one simple baseline from literature (e.g., train a basic GAN on our data) as a sanity check that we can get similar results to reported (though with 10 subjects maybe their results not directly comparable). - **Go/No-Go Criteria:** If we find that someone *already* did topology-preserving 3T->7T and solved it (unlikely, but if a 2024 paper popped up with identical approach), then we'd need to pivot (No-Go on original plan). Or if initial experiments show the dataset is too limited (e.g., we try a quick U-Net and it cannot learn anything, meaning 10 subjects maybe too few even with augmentation), then we reconsider approach (maybe need semi-supervised or synthetic data to augment training). - **Common Mistakes to Avoid:** Jumping into coding the complex model without verifying basics. We should avoid spending time on fancy stuff if maybe a simpler approach fails due to data. So in this phase, ensure that at least a baseline can produce something (even if poor) – that means the data is usable and the task is learnable.

Phase 1: Preprocessing Pipeline Lock-Down (Month 2) - **Deliverables:** A working preprocessing script that takes raw data to final paired inputs (aligned, normalized patches), plus QA/QC report on all subjects. We'll have visualizations of alignment for each subject, and metrics like MRIQC to ensure all images are acceptable. - **Tasks:** Implement all steps from Section 5: BIDS parsing, applying FLIRT (if needed; but we likely just use aligned images as given), skull stripping (test a couple of methods and choose best by visual inspection), bias correction, intensity normalization. Generate patches and confirm target alignment. - Also set aside test subjects if doing a fixed split, or plan cross-val folds here. - **Go/No-Go Criteria:** If we cannot get well-aligned brain images (say FLIRT fails badly on one subject), we decide how to handle (maybe drop that subject or apply manual intervention). If skull stripping is problematic, try alternatives until resolved. We only proceed when we are confident the preprocessed data is high quality; otherwise, any model results will be garbage. If issues persist (like alignment too off), consider doing our own registration or contacting data authors for advice (though likely fine). - **Common Mistakes:** Overlooking small misalignments that could hamper training. Also, we must ensure that we don't inadvertently use 7T info in preprocessing beyond alignment (alignment uses 7T as reference which is fine). We should double-check that the "aligned" images provided indeed use only rigid transforms (which they said yes [38]). Mistake to avoid: forgetting to apply the same transforms to T2 or not using correct interpolation (leading to slight mismatches). - We'll run a quick check: take a patch from 3T and 7T and see if the structures line up. This phase ends when any random patch visually shows matching anatomy in input vs target (apart from resolution/contrast differences).

Phase 2: Baseline Model Implementation & Evaluation (Month 3-4) - **Deliverables:** One or two baseline models fully trained and evaluated. For instance, a 3D U-Net with L1 loss (and maybe a Pix2Pix GAN variant) with results on cross-validation. This includes saved model weights and quantitative metrics recorded. - **Tasks:** Write training code for a basic supervised model. Probably start with the simplest: 3D U-Net, L1+L2 loss, no fancy stuff. Train it on patches from 8 subjects, validate on 1, etc. See if it can learn (we expect yes, it will at least learn to copy the image because 3T and 7T are similar). - Evaluate baseline on test fold(s): compute PSNR, etc., and also see what it does wrong (maybe it's blurry or misses fine edges). - Possibly implement a PatchGAN baseline too (if time permits) or note to do it later if needed. - **Go/No-Go Criteria:** If the baseline completely fails (like output is identical to input, meaning network learned identity because mapping is too hard or data too little), that's a red flag. However, I'd expect some improvement in contrast at least. If baseline fails, maybe we'll need to incorporate adversarial or something even for base. But given literature, a CNN should do something. - If baseline is okay, we analyze its shortcomings to justify our next steps. E.g., we notice “hippocampal detail still not there” or “image is smooth.” That analysis will directly feed into demonstrating why our advanced method is needed. - If baseline already miraculously produces perfect images (very unlikely), then our job is harder to find novelty. But with 10 subjects, baseline won't be perfect – likely blurred or losing fine texture. - **Common Mistakes:** Not tuning the baseline at all. Even for baseline, do a bit of hyperparam tuning (like choose a learning rate that converges, etc.). Otherwise we might misjudge the difficulty. Also, ensure baseline is trained long enough, because with so few subjects it might overfit extremely – we need to monitor if it just memorized training (see if test PSNR is much lower than train). - At this stage, we also double-check augmentation approach with baseline: maybe try training baseline with and without augmentation to see if augmentation actually helps generalization. This also informs how we approach main model. - We'll freeze the baseline once it's performing decently (not necessarily optimum, but representative of prior methods). - We should also finalize what baseline we present (maybe just one main baseline or two) for the paper, but we can do that later.

Phase 3: Novel Model Development (Month 5-8) - **Deliverables:** The full proposed model implemented, trained, and tuned. At least initial version by month 6 for testing, refined by month 8. This includes code for diffusion or transformer, multi-task outputs, etc. Also intermediate outputs (for example, segmentation head outputs to verify training). - **Tasks:** This is the core model building: - Implement the conditional diffusion model (or the GAN if we pivot). If diffusion: define network architecture (likely reuse parts of baseline U-Net code but add conditioning, maybe incorporate attention layers). - Implement the multi-task segmentation branch. That means adding an output head and computing segmentation loss. This requires preparing ground truth segmentation labels for training (we have them from FS). - Integrate the loss functions: combine L1, perceptual, segmentation. Set up training loop with the possibility of staged training (so perhaps code should allow turning on/off certain loss terms over epochs). - If using diffusion: implement noise schedule, training sampler. Possibly start with an existing diffusion repo and adapt it to conditional mode. - If using GAN: implement discriminator and training alternating updates. - **Debugging training**: likely needed. We might first try training without segmentation loss to get baseline diffusion working, then add segmentation. Monitor if segmentation loss actually decreases, etc. - Gradually incorporate all components. - Tune hyperparameters: loss weights (maybe try a couple values on a small validation set to see which yields best compromise of image vs seg accuracy), learning rate (monitor if training stable). - We should also tune patch size or batch size if needed for performance vs memory. - Possibly by mid-phase, we have initial results that we can compare to baseline to see improvement. If something’s not working (e.g., segmentation loss causes weird artifacts), adjust accordingly (maybe segmentation branch needs more capacity or weight adjusting). - **Go/No-Go Criteria:** If by end of Month 6 we cannot get the advanced model to train stably (say diffusion is too slow or unstable on so few data), we need to consider Plan B, perhaps a simpler model with GAN + segmentation loss which might be easier to train. We should allocate time to try an alternative if diffusion model underperforms baseline (that would be surprising but possible if data too small for diffusion). - A decision point: if diffusion is training but images are too noisy or not clearly better than baseline by a certain number of epochs, we could try increasing training time or adjusting. If still not better, consider switching to a GAN approach or a two-stage approach (like train deterministic then refine with diffusion). - We must decide by ~Month 7 if we stick with diffusion or pivot, so we have time to finalize results by Month 8. - **Common Mistakes:** Over-complicating without testing incrementally. It's important to validate each piece: first, does adding segmentation loss actually improve segmentation on val? Does adding perceptual loss actually sharpen images or just add noise? We'll use ablation mindset during training to get each piece right. - Another risk is spending too long chasing minimal improvements in hyperparameters. We should set a reasonable metric goal (like "achieve segmentation Dice > 0.95 for key structures and SSIM improvement over baseline by >0.02") and not over-tune beyond that for submission. - Also ensure we don't introduce a bug that uses test info or misaligns data, which can artificially inflate metrics. For example, if by mistake a segmentation label from test got into training, that's leakage. We will strictly separate data in code (and maybe assert that no test subject index is in training loader). - By end of Phase 3, we aim to have a model that clearly outperforms baseline in the ways we care about (especially anatomy preservation). We should have preliminary tables and images by this point, to confirm we have something publishable.

Phase 4: Comprehensive Evaluation & Ablation (Month 9) - **Deliverables:** Final evaluation results: all metrics computed, tables drafted, figures generated. Also ablation study results. Essentially all numbers and plots that will go into the paper. - **Tasks:** - Run the final model on cross-validation or test set and compute metrics as described in Section 8. - Compute baseline metrics for comparison if not done already. - Compute segmentation-based metrics: we might need to run FreeSurfer on synthetic images to get an "automatic" segmentation for evaluation (though we have GT from real 7T, one approach is to measure differences between FS on synthetic vs FS on real, which requires running FS on synthetic images). This can be time-consuming (FreeSurfer hours per subject). If too slow, perhaps use our segmentation head's output as segmentation for synthetic and compare to FS on real which we have as label. But using FS on synthetic might be a nice sanity check to mimic actual usage. If time, we do it for final results. - Collate those results (maybe make nice plots). - Perform statistical tests (use Python or R to do t-tests or wilcoxon, get p-values). - Create visual comparisons: pick a representative subject and make figure. Or possibly for multiple subjects, but likely one or two figures suffice. - Conduct ablation experiments: e.g., train model with $\lambda_{seg}=0$ (no topo loss) and evaluate the same metrics. That likely requires another training run or using an earlier checkpoint. We should allocate maybe a week or so to do these ablations (they might train faster if parts disabled). - If multi-modal input is an ablation, we run model with just T1 to see difference. - Possibly test model on a slightly out-of-distribution sample (maybe a 3T from another dataset if available) to see how it generalizes; not required but if we have ADNI 3T, run our model on it to show a qualitative result (could add in discussion). - Ensure all evaluation code is correct (no train data in test metrics etc.). - **Go/No-Go Criteria:** At this stage, it's more about completeness. If evaluation reveals a serious flaw (like our method improved some metrics but maybe made some others worse, e.g., if our method slightly blurs images relative to a GAN baseline which had sharper edges – could be a concern), we need to decide how to present it or if an adjustment needed. For instance, if we found that adding segmentation loss slightly reduced SSIM because it prioritized structure over texture, we might be okay with that and explain it. But if a baseline is better in an important metric (say baseline GAN had higher SSIM but we have better dice), we must be ready to justify focusing on our strengths. That's not a no-go, but something to handle in discussion. - Ideally, by now, we have all positives or at least a clear trade-off story. - **Common Mistakes:** Rushing through evaluation and making an error. E.g., using a non-brain region in metric calculations (like background can skew MSE – better to evaluate within brain mask). Or calculating metrics at different resolution incorrectly (but our images are same resolution so fine). - We must double-check each result's consistency. For instance, does the reported PSNR match what one would expect from visual difference? If something is off, find out why. - Also, not forgetting to evaluate on the actual independent test set if we kept one. If cross-val, we present average but perhaps also mention range or worst-case to be honest. - Another potential oversight: not evaluating inference speed or mentioning it. Reviewers might ask if method is practical. We should at least note how long it takes per volume (e.g., "synthesis of one volume takes 2 minutes on GPU"). We can get that from our code and include it (maybe in discussion, not main results). - By end of Phase 4, we have all the raw material (numbers, images) needed for writing the paper draft.

Phase 5: Writing, Feedback, and Submission (Month 10-12) - **Deliverables:** Complete manuscript ready for submission to chosen venue; possibly a preprint on arXiv for visibility. All accompanying files (code, supplemental documents, etc.). - **Tasks:** - Write the paper sections, as detailed in section 10. We'll allocate time to writing and figures. Possibly produce initial draft by mid-month 10. - Internally review the draft, have colleagues or mentors read it if possible to get feedback (especially checking if claims are clear and supported). - Revise accordingly. Pay attention to clarity, conciseness (NeuroImage allows decent length, but still be organized). - Prepare figures carefully: high resolution, properly labeled. E.g., a figure showing image comparisons should have arrows or circles highlighting important differences so reviewers don't miss them. - Prepare supplementary materials: e.g., more example images, more detailed tables, code snippet, anything that didn't fit in main text. - Write a cover letter to editors emphasizing why this work is important and suitable for the journal (mention how it leverages a new open dataset, solves a known limitation, etc.). - Ensure all references are in order (we have many from our search). - **Go/No-Go Criteria:** By now we definitely go to submission unless something disastrous was found (like a mistake in evaluation that undermines results – then fix it and possibly retrain if necessary). - Choose journal and format accordingly (NeuroImage has specific sections and a word count ~7000 typically; MedIA might be similar). - Possibly consider a fallback if journal rejects: plan to submit to another (like if NeuroImage rejects after review, go to TMI or MedIA, etc. But hopefully not needed). - **Common Mistakes:** - Not addressing all reviewer types in writing. We should ensure the paper has something for both technical and clinical readers (NeuroImage has both). We include technical details for method folks and also clear explanation of what it means for neuroscience/AD for clinician readers. - Avoid making unsubstantiated claims (like "This will revolutionize AD diagnosis" is too strong – better to say "This approach may assist in earlier detection..."). - Overloading the paper with too many experiments: we should keep focus on main message; extra tests can go to supplement. - Typos or grammar – need thorough proofread to appear professional. - In references, ensure to cite all relevant recent work including those by likely reviewers if applicable (just to show we know the field). For instance, if someone on MICCAI 2022 did something related, cite them to avoid them feeling ignored. - Code release: prepare repository and perhaps include link in the paper (some journals allow referencing code, or we say "will be released upon acceptance" if not comfortable releasing earlier). - We should also practice explaining the work verbally, anticipating questions (like a defense) – useful if presenting at a conference or if reviewers ask for clarifications.

Timeline Summary: - Months 1-2: Problem refinement and preprocessing – **foundation is set**. - Month 3-4: Baseline experiments – **feasibility check and performance gap identified**. - Month 5-8: Core model dev – **innovation implemented and optimized**. - Month 9: Evaluation – **evidence gathered**. - Month 10-12: Writing and polishing – **story finalized and communicated**.

We'll maintain a Gantt chart or task list to ensure we don't slip on any part. Although we have "no strict deadline", we impose one to keep momentum – say aim to submit to NeuroImage's special issue or something by end of year.

At each phase's end, we'll have a review meeting: - Phase 1 review: data quality and any necessary adjustments. - Phase 2 review: baseline performance and confirm need for next steps (if baseline already did extremely well, we reconsider how to differentiate). - Phase 3 review (mid): check if new model is trending better than baseline. If not, adjust approach quickly. - Phase 4 review: do results support our claims? If not, either adjust claims or run additional experiments to support. - Phase 5: maybe get an external reviewer (friendly colleague) to critique the draft before submission to catch any oversight.

Common execution pitfalls: - Delays in training (diffusion could take long). We should use a good GPU and optimize code (maybe use mixed precision to speed up). - Possibly find we need more data earlier than expected; if so, consider data augmentation by generating synthetic paired data: e.g. down-sample 7T to simulate 3T, then try to recover (could artificially enlarge dataset). Actually, that is an idea: we can augment training by taking each 7T image, degrade it (downsample + add noise) to create additional pseudo-3T, then train model to recover original 7T. That uses unpaired 7T data as well. If needed, that could be done to strengthen training. This is not exactly 3T->7T (since 7T->7T identity basically, but with degrade; still, it might teach network some high-frequency patterns). We'll keep it as a trick if model underfits (with 10, might underfit or actually more likely overfit).

Given all steps, we are quite comprehensive in planning. Now, we have to maintain discipline in following these steps and not skipping validation at each go/no-go. If something fails, better to catch in earlier phase (like data issues in Phase 1) than realize in Phase 4 that outputs are misaligned due to a data bug.

By Phase 5, we should have a strong story and evidence to back it, making for a solid Q1 submission.

12. Brutal Reality Check

Finally, we need to honestly assess: is this research idea worth pursuing now, and what is the most defensible angle?

Is it worth pursuing now (2025)? - **Yes, conditionally.** The idea of 3T-to-7T MRI enhancement remains relevant and not fully solved [7]. The availability of a new public dataset [37] and the emergence of new generative techniques (diffusion) make it timely to tackle it with fresh approaches. There's also high clinical interest in leveraging 7T benefits without 7T scanners, which gives the work significance. Reviewers and readers will appreciate methods that push image quality boundaries, especially with a clear application like AD research. - However, we must acknowledge risk: the dataset is very small. This inherently limits how far we can go in terms of generalizability claims. If we execute perfectly, we get a great proof-of-concept that might inspire further study, but some might say it needs more subjects or multi-center data to be truly convincing for clinical translation. This could be addressed by focusing our claims carefully (we present it as a methodological advance and initial validation, not a deployment-ready solution). - Provided we clearly demonstrate improved anatomical fidelity and link to AD markers, the paper should be strong enough for a top journal. If we were only showing marginal denoising or resolution improvement, it might not be enough novelty. But our emphasis on topology and clinical metrics is the key differentiator that can elevate it from incremental to significant.

Single Most Defensible Angle: If we had to choose one angle to defend under scrutiny, it is: - **Anatomical/topological fidelity in synthetic MRI** as evidenced by segmentation and structural metrics being essentially identical to real 7T. This is our strongest claim and contribution. We should hammer that: "Unlike prior works that focused solely on visual similarity, our approach guarantees that no new anatomically incorrect structures are introduced and no real structures are lost – a crucial requirement for using synthetic images in clinical analysis [32]." This angle is defensible because we will have quantifiable proof (Dice ~0.98 for segmentation, etc.). It's hard for a reviewer to refute a method that clearly does what it says (preserve anatomy) and improves things like measurement accuracy. Even if someone is not excited by diffusion or specific method, they cannot deny the value of not hallucinating data in medical imaging. - So, if nothing else, we emphasize: *our method makes synthetic 7T a reliable proxy for real 7T in terms of anatomical detail.* That's a new level of validation that others didn't provide.

If the idea was not worth it (hypothetically, or if we discovered issues): We consider alternatives or pivots: - **Pivot 1:** If we decided direct 3T->7T image synthesis is too risky or saturated, we could pivot to a slightly reframed problem: e.g., *"Harmonization of 3T and 7T data via a generative model"*. That is, instead of strictly super-res, make a model that can take a 3T and output a 7T or vice versa, focusing on cross-field *consistency.* This could be framed as domain adaptation/harmonization (a hot topic, as multi-site integration is important). However, that's quite close to what we do, just different spin. - **Pivot 2:** Use 7T images as teacher to improve analysis on 3T without actually generating images. For example, train a model to directly predict segmentation or AD risk from 3T with help of 7T supervision (like a knowledge distillation: use 7T to get better labels or features). This would skip image synthesis and go to the task. It's a different approach: rather than making a synthetic 7T, we make a better classifier that implicitly uses 7T information. This could be strong if our ultimate goal is AD detection. The advantage: one could avoid potential image artifacts and focus on metrics directly. The disadvantage: it's less visually demonstrative and more narrow (just solving classification maybe). - If our project was failing to produce good images, this could salvage it: we could show that using 7T supervision (via a teacher network or so) on 3T data improves AD classification by X%. That would still be publishable, though a different paper (foundation models and distillation approach). - But that pivot loses the "synthetic image" angle which is the original premise. It becomes more of a machine learning approach to achieve the same end (better diagnosis). - **Pivot 3:** If small-N is a killer issue, pivot to incorporate more data, e.g., *semi-supervised training using unpaired 7T.* The PLOS one did that [45]. We could pull a similar trick: find some unpaired 7T volumes (perhaps the dataset of 20 by Li et al. that PLOS used [44]), include them with cycle or diffusion unpaired training. That's not a conceptual pivot, just a method extension if needed. It's more work but could strengthen results and show our method can utilize extra unpaired data (foundation model style). - **Pivot 4 (if completely abandoning 3T->7T):** If we found that early AD detection is the goal and 3T->7T is too indirect, we might pivot to *analyzing 7T images we have for subtle markers and then develop a 3T tool to predict those markers.* For example, use the 7T to measure something (like hippocampal microstructure via T2*), then train a network to predict that measure from 3T. That's somewhat like what GAN-MAT did for microstructure from T1 [26]. But our unique twist could be topology still: ensure the predicted maps are topologically consistent maybe. This is a major pivot away from image generation to more of a regression of measurements. It's interesting but not as visually appealing as generating whole 7T images.

Given our current plan, such drastic pivots aren't needed unless results are very disappointing. But being prepared is good.

Brutal Honesty: - We must accept that with 10 subjects, our results are a proof-of-concept. Reviewers might say "Nice, but will this work on population scale?" We will proactively state that further validation on larger cohorts is needed. - There's also the risk that some might say "Why do we need 7T-like images if segmentation can be done on 3T reasonably?" We will highlight specific advantages (for example, FreeSurfer sometimes fails in hippocampal subfields on 3T, but on 7T it's better [57]; our synthetic could reduce such failures). - Another reality: The AD angle in our specific data is not directly proven because no AD cases. We need to be careful not to over-promise. The best we can do is show that any metric that correlates with AD (like hippocampal volume or cortical thickness in entorhinal) is more accurately measured. That implies potential to detect smaller changes. But to truly test detection, one would need mild AD scans. We'll mention that as future work.

If the idea were not worth it, alternate strong pivots: - Perhaps *7T to 3T image degradation modeling* as a physics-informed cycle: Could invert problem and say we generate synthetic 3T from 7T (which we can do by adding noise/downsampling), then enhance back, ensuring cycle consistency. But that's just an approach for training essentially. - *Using 7T for enhanced structural connectivity or other analysis.* Not directly relevant.

Given what we know, the idea is worth pursuing – with caution about not overstating clinical proof. The most defensible novelty is that we are ensuring synthetic images are not just pretty but **quantitatively trustworthy** for anatomical analysis. That will likely be the line that sells the paper to reviewers.

We will be brutally honest in the paper about limitations so reviewers see we are self-aware: e.g., "Our model does not magically create entirely new information; if a pathological feature is totally absent at 3T, it won't appear at synthetic 7T. However, it can enhance subtle signals that were buried in noise at 3T, as long as they were present in some form. Thus, it's an aid to radiological analysis, not a replacement for actual high-field imaging when that is available." This kind of honest statement builds credibility.

In conclusion, we will proceed with the plan, focusing on the **anatomical fidelity** angle as the selling point. If any pivot is needed mid-way, it would likely be to incorporate additional unpaired data to strengthen training, rather than change the fundamental goal.

We will now move forward, fully aware of the challenges but with a solid plan to address them at each stage. The outcome, if executed well, will be a **comprehensive research blueprint** that stands up to rigorous review and advances the field of medical image enhancement.

---

1 2 14 51 57 Seven Tesla MRI in Alzheimer's disease research: State of the art and future directions: A narrative review
https://www.aimspress.com/article/doi/10.3934/Neuroscience.2023030?viewType=HTML

3 4 5 6 7 9 10 15 23 34 36 37 38 39 40 41 52 54 56 A paired dataset of T1- and T2-weighted MRI at 3 Tesla and 7 Tesla - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC10374655/

8 13 25 53 Synthetizing SWI from 3T to 7T by generative diffusion network for deep medullary veins visualization - ScienceDirect
https://www.sciencedirect.com/science/article/pii/S1053811925004781

11 12 16 17 18 19 20 21 22 Synthesized 7T MRI from 3T MRI via Deep Learning in Spatial and Wavelet Domains - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC7237331/

24 28 31 32 33 42 43 44 45 50 Semi-supervised synthesis of 7T MRI from 3T using 3D FR-U-Net with anatomical segmentation consistency assessment - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC12591452/

26 GAN-MAT: Generative adversarial network-based microstructural profile covariance analysis toolbox - ScienceDirect
https://www.sciencedirect.com/science/article/pii/S1053811924000909

27 35 47 ecva.net
https://www.ecva.net/papers/eccv_2020/papers_ECCV/papers/123480120.pdf

29 30 46 48 49 Vascular-Aware Multimodal MR–PET Reconstruction for Early Stroke Detection: A Physics-Informed, Topology-Preserving, Adversarial Super-Resolution Framework | MDPI
https://www.mdpi.com/2076-3417/15/22/12186

[55] HuXiaoling/awesome-topology-driven-image-analysis - GitHub
https://github.com/HuXiaoling/awesome-topology-driven-image-analysis